New

101 Days
Self-Preparation module

2nd edition

ENGLISH Medium

SSC MATHS

CHAPTER WISE TYPE WISE

6500+ TCS MCQ

- Complete Coverage for 2025 Exams
- Smart Approach & Short Tricks
- With Best Explanations

Useful for

SSC CGL (Tier-I & II), CHSL (Tier-I & II), CPO, Selection Post, GD, MTS & other Govt. Exams

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(Excise Inspector)

SELECTED हैं SELECTION दिलाएँगे



MASTER SSC MATHS LIKE A TOPPER

Your Ultimate Guide to Using This Book Effectively

Welcome to a power-packed Maths resource designed exclusively for SSC aspirants! This book isn't just a collection of past year questions-it's a strategic tool crafted to help you master every concept in a smart, structured, and exam-oriented way.

Why This Book is a Game-Changer?

- ✅ Complete Coverage of SSC Exams - Chapter-wise compilation of all questions asked in the last 5 years across SSC CGL, CHSL, CPO, MTS, and more!
- ✅ Type-wise & Level-wise Arrangement - Focus on one concept at a time and progress from basics to advanced effortlessly.
- ✅ Smart & Time-Saving Solutions - Master multiple solving techniques (traditional, shortcut, and conceptual) for every question.
- ✅ Designed for Targeted Practice - Learn a concept in class? Jump straight to practice only that type from this book!

How to Use This Book for Maximum Success?

📖 **STEP 1: Learn the Concept First**
Before diving into questions, understand the fundamentals from your teacher, notes, or video lectures.

🎯 **STEP 2: Targeted Concept Practice**
Thanks to the type-wise arrangement, you can instantly practice only the relevant question types, reinforcing your learning.

📊 **STEP 3: Level-Up Gradually**
Move step by step from basic to high-difficulty questions. This systematic progression boosts confidence and accuracy.

⚡ **STEP 4: Master Shortcuts & Multiple Methods**
For every question, compare different solving techniques-pick the fastest one to optimize your speed in exams.

🔄 **STEP 5: Revise & Track Your Progress**
Use the previous year questions (PYQs) as a self-evaluation tool. Reattempt difficult questions and refine your approach.

Pro-Tips to Ace SSC Maths Using This Book

- 🔥 Smart Work > Hard Work - Don't just practice, analyze solutions to find the best method.
- 🔥 Time Yourself - Try to solve questions within exam-time constraints for real-time improvement.
- 🔥 Make Notes of Tricks & Formulas - A quick revision sheet will be your best friend before the exam.
- 🔥 Solve Questions Daily - Even 15-20 quality questions per day can skyrocket your preparation.

🚀 Let's Ace SSC Maths Together!

With this book as your trusted guide, be assured of concept clarity, speed, and accuracy-the three pillars of SSC exam success!

🔷 Stay Consistent. Stay Focused. Stay Ahead. 🔷
🎯 All the Best! Your Success Story Starts Here. 🎯

Reverse Calendar Technique

101 days Self-Preparation module

101	100	99	98	97	96	95	94
93	92	91	90	89	88	87	86
85	84	83	82	81	80	79	78
77	76	75	74	73	72	71	70
69	68	67	66	65	64	63	62
61	60	59	58	57	56	55	54
53	52	51	50	49	48	47	46
45	44	43	42	41	40	39	38
37	36	35	34	33	32	31	30
29	28	27	26	25	24	23	22
21	20	19	18	17	16	15	14
13	12	11	10	9	8	7	6
5	4	3	2	1			

• INDEX •

S. NO.	CHAPTER	NAME OF TYPE		NO. OF QUES.	PAGE NO.	TIME TABLE
1.	Percentage	TYPE-I	Basic Questions	42	01 - 02	DAY 1-7
		TYPE-II	AB Rule	06	02 - 02	
		TYPE-III	Perimeter, Area & Volume	24	03 - 03	
		TYPE-IV	Successive Percentage Change	31	03 - 05	
		TYPE-V	Ratio Method	15	05 - 06	
		TYPE-VI	Reverse Case	02	06 - 06	
		TYPE-VII	Series Concept	42	06 - 07	
		TYPE-VIII	Price-Consumption & Expenditure	14	08 - 08	
		TYPE-IX	Income-Expenditure & Savings	02	08 - 08	
		TYPE-X	Based on Alligation	39	08 - 10	
		TYPE-XI	Based on Passing/Failure in an Exam	05	10 - 10	
		TYPE-XII	Based on Election	88	10 - 14	
		TYPE-XIII	Venn Diagram	08	15 - 15	
		TYPE-XIV	Commission	07	15 - 15	
		TYPE-XV	Income Tax	06	15 - 15	
		TYPE-XVI	Miscellaneous Questions	85	16 - 20	
			Answer Key	–	21 - 21	
			Solutions	–	22 - 46	
			Total	416		
2.	Profit & Loss	TYPE-I	Basic Questions	35	47 - 48	DAY 8-11
		TYPE-II	Difference Between the SP & CP	06	48 - 48	
		TYPE-III	CP of (X) = SP of (Y)	12	48 - 49	
		TYPE-IV	Profit/Loss = CP/SP of Some Articles	06	49 - 49	
		TYPE-V	Profit/Loss on Selling Price	09	49 - 49	
		TYPE-VI	Butterfly Concept	08	49 - 50	
		TYPE-VII	Number of Articles	08	50 - 50	
		TYPE-VIII	Average Concept (When CP is Same)	16	50 - 51	
		TYPE-IX	When SP is Same	15	51 - 51	
		TYPE-X	CP of 1st Article = SP of 2nd Article	01	51 - 51	
		TYPE-XI	Dishonest Shopkeeper	45	52 - 54	
		TYPE-XII	Char Minar Concept	12	54 - 54	
		TYPE-XIII	Based on Alligation	15	54 - 55	





		TYPE-XIV	Questions based on $P_1/Q_1 \times (100 \pm P_1/L_1)\% = P_2/Q_2 \times (100 \pm P_2/L_2)\%$	07	55 - 55	
		TYPE-XV	Miscellaneous Questions	58	55 - 58	
			Answer Key	–	59 - 59	
			Solutions	–	59 - 74	
			Total	253		
3.	Discount	TYPE-I	Basic Questions	16	75 - 75	DAY 12-14
		TYPE-II	Buy X Get Y Free	06	75 - 76	
		TYPE-III	Equivalent Discount of Two Successive Discounts	27	76 - 77	
		TYPE-IV	Equivalent Discount of Three Successive Discounts	06	77 - 77	
		TYPE-V	Series Concept	45	77 - 79	
		TYPE-VI	Mumtaz Concept	35	79 - 81	
		TYPE-VII	Miscellaneous Questions	70	81 - 84	
			Answer Key	–	85 - 85	
			Solutions	–	85 - 98	
			Total	205		
4.	Simple Interest	TYPE-I	Basic Questions	97	99-102	DAY 15-18
		TYPE-II	Concept Of Time	21	102-103	
		TYPE-III	Rate is X% Higher or Lower	12	103-104	
		TYPE-IV	Munni Method	12	104-104	
		TYPE-V	Based on Alligation	12	104-105	
		TYPE-VI	Concept of Total SI	08	105-105	
		TYPE-VII	Concept of Equal SI	07	105-106	
		TYPE-VIII	Miscellaneous Questions	56	106-108	
		TYPE-IX	Installment	34	108-110	
			Answer Key	–	110-111	
			Solutions	–	111-123	
			Total	259		
5.	Compound Interest	TYPE-I	Based on 2 Cycle	14	124-124	DAY 19-20
		TYPE-II	Based on 3 & 4 Cycle	25	124-125	
		TYPE-III	When Time is Given in Fraction	04	125-125	
		TYPE-IV	Based on Difference of CI for 2 & 3 Cycle	12	126-126	
		TYPE-V	Combination of CI and SI both	17	126-127	
		TYPE-VI	How to Find Rate	07	127-127	
		TYPE-VII	How to Find Time	08	127-127	
		TYPE-VIII	Munni Method	09	128-128	
		TYPE-IX	Principal Becomes N Time After T Years	03	128-128	
		TYPE-X	Miscellaneous Questions	14	128-129	
		TYPE-XI	Installment	19	129-130	
			Answer Key	–	130-130	
			Solutions	–	131-138	
			Total	132		

6.	Ratio	TYPE-I	Basic Questions	57	139-140	DAY 21-22
		TYPE-II	Word Problems Based on Basic Concepts	34	140-142	
		TYPE-III	Based on Addition/Subtraction	16	142-143	
		TYPE-IV	Based on Coins	14	143-143	
		TYPE-V	Income Expenditure & Savings	14	143-144	
		TYPE-VI	Miscellaneous Questions	18	144-145	
			Answer Key	–	145-145	
			Solutions	–	145-153	
			Total	153		
7.	Proportion	TYPE-I	When Three Numbers are Given	27	154-154	DAY 23-24
		TYPE-II	When Two Numbers are Given	40	154-156	
		TYPE-III	Proportion After Addition/Subtraction	05	156-156	
		TYPE-IV	Miscellaneous Questions	18	156-157	
			Answer Key	–	157-157	
			Solutions	–	157-160	
			Total	90		
8.	Age	TYPE-I	Basic Questions	22	161-161	DAY 25-25
		TYPE-II	Based on Average Age	07	162-162	
		TYPE-III	Based on Difference of Age	17	162-163	
		TYPE-IV	Miscellaneous Questions	04	163-163	
			Answer Key	–	163-163	
			Solutions	–	163-166	
			Total	50		
9.	Partnership	TYPE-I	Basic Questions	16	167-167	DAY 26-26
		TYPE-II	When Someone Joined or Left	07	167-168	
		TYPE-III	Increase/Decrease in Capital	07	168-168	
		TYPE-IV	Based on Donation/Tax	04	168-168	
		TYPE-V	Based on Distribution	02	168-168	
		TYPE-VI	Based on Management	02	169-169	
		TYPE-VII	Miscellaneous Questions	02	169-169	
			Answer Key	–	169-169	
			Solutions	–	169-171	
			Total	40		
10.	Mixture & Alligation	TYPE-I	Based on Addition or Removal	18	172-172	DAY 27-27
		TYPE-II	Concept of Replacement	11	172-173	
		TYPE-III	Mixing Different Mixtures to Form A Single Mixture	33	173-175	
		TYPE-IV	Based on Alligation	11	175-175	
		TYPE-V	Based on Profit/Loss	10	175-175	
		TYPE-VI	Miscellaneous Questions	09	176-176	
			Answer Key	–	176-176	
			Solutions	–	177-183	
			Total	92		

11.	Average	TYPE-I	Basic Questions	32	184-185	DAY 28-31
		TYPE-II	Based on Consecutive Natural Numbers	27	185-186	
		TYPE-III	Based on Weighted Average	28	186-187	
		TYPE-IV	Based on Inclusion	18	187-188	
		TYPE-V	Based on Exclusion	08	188-188	
		TYPE-VI	Based on Replacement	14	188-189	
		TYPE-VII	Based on Wrongly Entered Data	08	189-189	
		TYPE-VIII	Based on Alligation	12	189-190	
		TYPE-IX	Based on Cricket	05	190-190	
		TYPE-X	Based on Age	09	190-191	
		TYPE-XI	Based on Numbers	18	191-191	
		TYPE-XII	Miscellaneous Questions	39	192-193	
			Answer Key	–	194-194	
			Solutions	–	194-206	
			Total	218		
12.	Time & Work	TYPE-I	Based on Two Variables	39	207-208	DAY 32-35
		TYPE-II	Based on Three Variables	46	209-211	
		TYPE-III	Concept of Alternate Days	41	211-213	
		TYPE-IV	Concept of Lefting The Work	21	213-215	
		TYPE-V	Concept of Efficiency	46	215-217	
		TYPE-VI	MDH Concept	27	217-218	
		TYPE-VII	Concept of MWC	28	218-219	
		TYPE-VIII	Concept of Contractor	02	219-219	
		TYPE-IX	Concept of AP in Time & Work	03	219-220	
		TYPE-X	Miscellaneous Questions	05	220-220	
		TYPE-XI	Work & Wages	18	220-221	
			Answer Key	–	221-221	
			Solutions	–	222-239	
			Total	276		
13.	Pipe & Cistern	TYPE-I	Completely Filling or Emptying	23	240-241	DAY 36-36
		TYPE-II	Filling & Emptying Simultaneously	21	241-242	
		TYPE-III	Taps Opened Alternately	04	242-242	
		TYPE-IV	To Find the Capacity of Tank	05	242-242	
		TYPE-V	Leakage in Tank	10	242-243	
		TYPE-VI	When 1 or 2 Taps are Closed During The Filling or Emptying a Tank	25	243-244	
			Answer Key	–	245-245	
			Solutions	–	245-251	
			Total	88		

14.	Time & Distance	TYPE-I	Basic Questions	24	252-253	DAY 37-40
		TYPE-II	When Speed is Increased/ Decreased	11	253-253	
		TYPE-III	When Distance is Constant and Total Time is Given	07	253-253	
		TYPE-IV	When Distance is Constant and Difference of Time is Given	13	254-254	
		TYPE-V	Average Speed	60	254-257	
		TYPE-VI	Average Speed With Stoppage	05	257-257	
		TYPE-VII	Radha Concept	05	257-258	
		TYPE-VIII	Based on Police and Thief	50	258-260	
		TYPE-IX	Relative Speed	37	260-262	
		TYPE-X	Miscellaneous Questions	37	262-264	
			Answer Key	–	264-265	
			Solutions	–	265-279	
			Total	249		
15.	Race & Circular Motion	TYPE-I	Concept of Linear Race	39	280-281	DAY 41-41
		TYPE-II	Concept of Circular Race	39	281-283	
			Answer Key	–	283-283	
			Solutions	–	284-288	
			Total	78		
16.	Train	TYPE-I	Train Passing a Telegram Pole, Bridge or Platform	07	289-289	DAY 42-42
		TYPE-II	Two Trains Crossing Each other in Opposite Direction	16	289-290	
		TYPE-III	Two Trains Crossing Each other in Same Direction	14	290-290	
		TYPE-IV	When a Train Crosses 1 or 2 Running Persons	07	290-291	
		TYPE-V	When a Train Crosses a Person Sitting Inside Another Train	04	291-291	
		TYPE-VI	Miscellaneous Questions	03	291-291	
			Answer Key	–	291-291	
			Solutions	–	291-294	
			Total	51		
17.	Boat & Stream	TYPE-I	Basic Questions	35	295-296	DAY 43-43
		TYPE-II	Base on: $B/S = (T_1+T_2) / (T_2-T_1)$	06	296-296	
		TYPE-III	Base on: $T = 2Dx/(x^2 - y^2)$	04	297-297	
		TYPE-IV	Base on: $D = T(B^2-S^2/2B)$	10	297-297	
		TYPE-V	UV Concept	14	297-298	
		TYPE-VI	Miscellaneous Questions	08	298-298	
			Answer Key	–	298-298	
			Solutions	–	299-304	
			Total	77		

18.	Number System	TYPE-I	Classification of Numbers	21	305-305	DAY 44-49
		TYPE-II	Unit Digit	12	305-306	
		TYPE-III	Factor	16	306-306	
		TYPE-IV	Divisibility	179	306-311	
		TYPE-V	Remainder	88	311-314	
		TYPE-VI	Miscellaneous Questions	44	314-316	
			Answer Key	–	316-316	
			Solutions	–	317-331	
			Total	360		
19.	LCM & HCF	TYPE-I	Basic Questions	20	332-332	DAY 50-50
		TYPE-II	When The Ratio of Numbers is Given	08	332-333	
		TYPE-III	1st Number × 2nd Number = L.C.M. × H.C.F.	20	333-333	
		TYPE-IV	Application of L.C.M.	17	333-334	
		TYPE-V	Application of H.C.F.	08	334-334	
		TYPE-VI	Miscellaneous Questions	11	334-335	
			Answer Key	–	335-335	
			Solutions	–	335-338	
			Total	84		
20.	Simplification	TYPE-I	Based on BODMAS Rule	173	339-345	DAY 51-54
		TYPE-II	Based on Algebraic Formulae	42	346-347	
		TYPE-III	Comparison of Fraction	04	347-347	
		TYPE-IV	Ladder Fraction	05	347-348	
		TYPE-V	Based on Bar	03	348-348	
		TYPE-VI	Continuous Fraction Series	06	348-348	
		TYPE-VII	Concept of AP & GP	03	348-348	
			SURDS & INDICES			
		TYPE-VIII	Basic Questions	07	348-348	
		TYPE-IX	To Find Smallest/Largest	05	349-349	
		TYPE-X	Infinite Series	08	349-349	
		TYPE-XI	Square Root in Denominator	10	349-349	
		TYPE-XII	Simplification of Powers	03	349-349	
			Answer Key	–	350-350	
			Solutions	–	351-363	
			Total	269		
21.	Algebra	TYPE-I	Inverse Function (Basic Questions)	51	364-366	DAY 56-61
		TYPE-II	Inverse Function (Quadratic Equation)	24	366-367	
		TYPE-III	Inverse Function (Reverse Case)	30	367-368	
		TYPE-IV	Inverse Function (Concept of Bridging)	27	368-369	
		TYPE-V	Inverse Function (Unequal Co-efficient)	07	369-369	
		TYPE-VI	Inverse Function (Special Cases)	19	369-370	
		TYPE-VII	Inverse Function (Miscellaneous Questions)	16	370-370	

		TYPE-VIII	Square Formulae (Two Variables)	42	370-372	
		TYPE-IX	Cube Formulae (Two Variables)	40	362-373	
		TYPE-X	Concept of Perfect Square	07	373-373	
		TYPE-XI	Square Formulae (Three Variables)	14	373-374	
		TYPE-XII	Cube Formulae (Three Variables)	43	374-375	
		TYPE-XIII	Concept of Value Putting	24	375-376	
		TYPE-XIV	Miscellaneous Questions	57	376-378	
			Answer Key	–	379-379	
			Solutions	–	380-401	
			Total	401		
22.	Trigonometry	TYPE-I	Basic Trigonometric Ratios	97	402-406	DAY 62-70
		TYPE-II	Values of Trigonometric Ratios	48	406-408	
		TYPE-III	Based on Quadrant	14	408-409	
		TYPE-IV(A)	If $\alpha+\beta=90^\circ \Rightarrow \alpha$ & β are Complementary	28	409-410	
		TYPE-IV(B)	If $\alpha+\beta=90^\circ \Rightarrow \alpha$ & β are Complementary	16	410-411	
		TYPE-V(A)	Trigonometric Identities (Basic Qns.)	96	411-414	
		TYPE-V(B)	Based on Algebraic Formulae	16	414-415	
		TYPE-V(C)	If $\sec\theta + \tan\theta = K$, Then $\sec\theta - \tan\theta = 1/K$, If $\operatorname{cosec}\theta - \cot\theta = K$, Then $\operatorname{cosec}\theta + \cot\theta = 1/K$	15	415-415	
		TYPE-V(D)	Based on Quadratic Equations	23	415-416	
		TYPE-V(E)	Trigonometric Identities (Adv. Qns.)	43	416-418	
		TYPE-V(F)	Bahubali Concept	11	418-418	
		TYPE-V(G)	Some Special Concepts	42	418-420	
		TYPE-VI	$a\sin\theta + b\cos\theta = x$, $a\cos\theta - b\sin\theta = y$, Then, $a^2 + b^2 = x^2 + y^2$	08	420-420	
		TYPE-VII	Concept of Value Putting	15	420-421	
		TYPE-VIII	Miscellaneous Questions	12	421-421	
		TYPE-IX(A)	Trigonometric Functions of 2A & 3A	15	422-422	
		TYPE-IX(B)	Sum & Difference of Angles of Trigonometric Ratios	18	422-423	
		TYPE-IX(C)	Transformation of Sum or Difference Into Product	04	423-423	
		TYPE-IX(D)	Transformation of Product Into Sum or Difference	04	423-423	
		TYPE-IX(E)	Maximum/Minimum Values of Trigonometric Functions	07	423-423	
			Answer Key	–	424-424	
			Solutions	–	425-461	
			Total	532		

23.	Height & Distance	TYPE-I	We have categorized this chapter based on figures, which is why we haven't assigned specific names to the types.	15	462-462	DAY 71-71
		TYPE-II		08	462-463	
		TYPE-III		05	463-463	
		TYPE-IV		09	463-463	
		TYPE-V		07	463-464	
			Answer Key	–	464-464	
			Solutions	–	464-468	
			Total	44		
24.	Geometry	TYPE-I (A)	Line & Angle	03	469-469	DAY 72-81
		TYPE-I (B)	Fundamental Properties of Triangle	72	469–471	
		TYPE-II	Congruency & Similarity	127	471–477	
		TYPE-III	Type of Triangles	71	477–480	
		TYPE-IV	Centers of Triangle	65	480–482	
		TYPE-V (A)	Quadrilaterals	21	482–483	
		TYPE-V (B)	Polygon	08	483–483	
		TYPE-VI	Circle	221	483–493	
		TYPE-VII	Miscellaneous Questions	34	493–495	
			Answer Key	–	495–496	
			Solutions	–	497–543	
			Total	622		
25.	Co-Ordinate Geometry	TYPE-I	Basic Questions	07	544–544	DAY 82-82
		TYPE-II	Distance Formula	02	544–544	
		TYPE-III	Mid Point Formula	01	544–544	
		TYPE-IV	Section Formula	03	544–544	
		TYPE-V	Slop of a Line	11	544–544	
		TYPE-VI	Point of Intersection of Two Lines	08	544–545	
		TYPE-VII	Angle Between Two Lines	01	545–545	
		TYPE-VIII	Distance of a Point From a Line	03	545–545	
		TYPE-IX	Triangle	13	545–545	
		TYPE-X	Mirror/Reflection	03	545–545	
		TYPE-XI	Circle	04	545–545	
		TYPE-XII	System of Equations	32	546–547	
			Answer Key	–	547–547	
			Solutions	–	547–552	
			Total	88		

26.	Mensuration -2D & 3D		Mensuration–2D			DAY 83-93
		TYPE-I (A)	Equilateral Triangle	16	553–553	
		TYPE-I (B)	Scalene Triangle	05	553–553	
		TYPE-I (C)	Isosceles Triangle	11	553–554	
		TYPE-I (D)	Right Angled Triangle	17	554–554	
		TYPE-II	Circle	45	554–556	
		TYPE-III	Sector of a Circle	42	556–558	
		TYPE-IV	Quadrilateral	05	558–558	
		TYPE-V	Rectangle	28	558–559	
		TYPE-VI	Square	23	559–560	
		TYPE-VII	Parallelogram	06	560–560	
		TYPE-VIII	Rhombus	14	560–561	
		TYPE-IX	Trapezium	05	561–561	
			Mensuration–3D			
		TYPE-I	Cube	23	561–562	
		TYPE-II	Cuboid	67	562–564	
		TYPE-III	Cylinder	73	564–567	
		TYPE-IV	Hollow Cylinder	05	567–567	
		TYPE-V	Cone	52	567–569	
		TYPE-VI	Frustum	02	569–569	
		TYPE-VII	Sphere	41	569–571	
		TYPE-VIII	Spherical Shell	03	571–571	
		TYPE-IX	Hemisphere	26	571–572	
		TYPE-X	Hemispherical Shell	06	572–572	
		TYPE-XI	Concept of Melting & Recasting	60	572–574	
		TYPE-XII	Concept of Digging & Earth Taken Out	02	574–574	
		TYPE-XIII	Filling a container by Another Shape Container	05	575–575	
		TYPE-XIV	Concept of Rise in Height	02	575–575	
		TYPE-XV	Combination of 3D Figures	19	575–576	
		TYPE-XVI	Prism	09	576–576	
		TYPE-XVII	Pyramid	10	576–577	
			Answer Key	–	578–579	
			Solutions	–	579–608	
			Total	622		

27.	Statistics	TYPE-I	Mean	19	609-610	DAY 94-94
		TYPE-II	Median	24	610-611	
		TYPE-III	Mode	14	611-612	
		TYPE-IV	Range	06	612-612	
		TYPE-V	Variance	07	612-613	
		TYPE-VI	Standard Deviation	12	613-614	
		TYPE-VII	Coefficient of Variation	02	614-614	
			Answer Key	–	614-614	
			Solutions	–	614-619	
			Total	84		
28.	Permutation & Combination	TYPE-I	Concept of Factorial	01	620-620	DAY 95-95
		TYPE-II	Permutation Formulae	15	620-621	
		TYPE-III	Permutations Formula With Repetition	11	621-621	
		TYPE-IV	Combination Formulae	14	621-622	
			Answer Key	–	622-622	
			Solutions	–	622-624	
			Total	41		
29.	Probability	TYPE-I	Basic Questions	09	625-626	DAY 96-97
		TYPE-II	Questions Based on Coins	09	626-626	
		TYPE-III	Questions Based on Dice	14	626-627	
		TYPE-IV	Questions Based on Playing Cards	20	627-628	
		TYPE-V	Questions Based on Ball/Marbles	22	628-629	
		TYPE-VI	Set Theory & Venn Diagram	16	630-631	
		TYPE-VII	Odds (In Favour & Against)	15	631-632	
		TYPE-VIII	Conditional Probability	20	632-633	
		TYPE-IX	Baye's Theorem	04	633-633	
		TYPE-X	Miscellaneous Questions	33	633-635	
			Answer Key	–	636-636	
			Solutions	–	636-644	
			Total	162		
30.	Data Interpretation	TYPE-I	Bar Graph	105	645-645	DAY 98-101
		TYPE-II	Tabular Chart	206		
		TYPE-III	Pie Chart	91		
		TYPE-IV	Line Graph	21		
			Answer Key	–		
			Solutions	–		
			Total	423		
			Grand Total	6459		

New

SSC MATHS

2nd edition

ENGLISH Medium

TYPE WISE

6500+ TCS MCQ

FREE

SIMPLE INTEREST

04 Chapter SIMPLE INTEREST

◆ TYPE – I ◆
BASIC QUESTIONS

1. Find the simple interest on ₹15,000 at the rate of 7% p.a. for 3 years.
SSC Phase XII 26/06/2024 (Shift-02)
(a) ₹3,000 (b) ₹3,100
(c) ₹3,150 (d) ₹3,120

2. A sum of ₹5,575 is invested at the rate of 8% simple interest per annum for 4 years. What will be the interest (in ₹) payable on maturity ?
SSC Phase XII 24/06/2024 (Shift-03)
(a) 1776 (b) 1784
(c) 1782 (d) 1788

3. Find the simple interest (in ₹) on ₹9,900 at 9% per annum in 3 years
SSC Phase XII 24/06/2024 (Shift-02)
(a) 2512 (b) 2683
(c) 2582 (d) 2673

4. A sum of ₹1,250 is lent on simple interest at a rate of $5\frac{1}{2}$% per annum for three years. Find the interest (in ₹)
SSC Phase XII 20/06/2024 (Shift-03)
(a) 210.25 (b) 200.00
(c) 215.75 (d) 206.25

5. How much simple interest will Rs.5,000 earn in 14 months at a 12% interest rate per annum?
SSC CHSL 11/07/2024 (Shift-04)
(a) ₹ 700 (b) ₹ 600
(c) ₹ 800 (d) ₹ 500

6. Find the simple interest on ₹ 8,000 at the rate of 10% per annum in 6 years.
SSC MTS 12/09/2023 (Shift- 02)
(a) ₹ 4,600 (b) ₹ 4,200
(c) ₹ 4,800 (d) ₹ 4,400

7. What sum of money (in ₹) will earn an interest of ₹ 1800 at the rate of 6% per annuun simple interest in five years?
SSC MTS 11/09/2023 (Shift-01)
(a) 5,000 (b) 18,000
(c) 6,000 (d) 8,000

8. Find the simple interest (in ₹) on ₹5,400 for 7 years at 4% per annum.
SSC MTS 06/09/2023 (Shift-01)
(a) 1,230 (b) 1,380
(c) 1,172 (d) 1,512

9. Find the simple interest on ₹1,35,000 at $6\frac{3}{4}$% per annum for 6 years.
SSC Phase XI 28/06/2023 (Shift-04)
(a) ₹54,575 (b) ₹54,675
(c) ₹54,475 (d) ₹54,875

10. How much simple interest will ₹6,000 earn in 21 months at 8% per annum?
SSC CGL 19/07/2023 (Shift-01)
(a) ₹750 (b) ₹880
(c) ₹620 (d) ₹840

11. How much simple interest will Rs.6,700 earn in 13 months at a 12% interest rate per annum?
SSC CGL 12/09/2024 (Shift-03)
(a) Rs.871 (b) Rs.889
(c) Rs.821 (d) Rs.813

12. Find the simple interest on ₹24,000 at 18% per annum for a period of 7 months
SSC Phase-XI 30/06/2023 (Shift-04)
(a) ₹ 2,525 (b) ₹ 2,520
(c) ₹ 2,530 (d) ₹ 2,535

13. The simple interest (in ₹ rounded off to the tens place) on ₹12300 from 22 May 1993 to 2 August 1993 (both days are included) at the rate of 12% per annum is:
SSC CHSL 09/08/2023 (Shift-04)
(a) 300 (b) 200
(c) 250 (d) 350

14. Manish invested a certain sum of money at 10% per annum simple interest. If he receives an interest of ₹72780 after 1 year, the sum (in ₹) he invested is:
SSC Phase XII 21/06/2024 (Shift-03)
(a) 728383 (b) 727812
(c) 727925 (d) 727800

15. Suresh invested a certain sum of money at 5% per annum simple interest. If he receives an interest of ₹69687 after 15 years, the sum (in ₹) he invested is:
SSC Phase XII 25/06/2024 (Shift-03)
(a) 92916 (b) 92880
(c) 92456 (d) 92530

16. An amount invested fetched a total simple interest of ₹4,050 at the rate of 9% per annum in 5 years. What is the amount invested (in ₹)?
SSC CHSL 09/07/2024 (Shift-03)
(a) 8,300 (b) 7,500
(c) 9,050 (d) 9,000

17. Suman paid ₹9,600 as interest on a loan she obtained 5 years ago with a simple interest rate of 16%. What was the amount of the loan she had taken?
SSC CGL 03/12/2022 (Shift-02)
(a) ₹13,250 (b) ₹12,500
(c) ₹12,000 (d) ₹11,750

18. Vinita obtained an amount of ₹23,340 as simple interest on a certain amount at 15% per annum after 8 years. What was the amount invested by Vinita?
SSC MTS 08/09/2023 (Shift-01)
(a) ₹ 17,490 (b) ₹ 20,450
(c) ₹ 19,450 (d) ₹ 18,650

19. A sum of simple interest at the rate of 5% per annum amount ₹4160 in 6 years. Find the sum.
SSC CHSL TIER-II (18/11/2024)
(a) ₹ 2800 (b) ₹ 2900
(c) ₹ 3200 (d) ₹ 3100

20. What principal would amount to ₹21,420 in 2 years at the rate of 9.5% p.a. simple interest?
SSC CGL 26/09/2024 (Shift-02)
(a) ₹ 11,273 (b) ₹ 12,000
(c) ₹ 18,000 (d) ₹ 16,000

21. A sum becomes ₹6,600 in 4 years at simple interest at a yearly interest rate of 5% per annum. What is the sum?
SSC CHSL 01/07/2024 (Shift-04)
(a) ₹ 6,000 (b) ₹ 4,400
(c) ₹ 5,500 (d) ₹ 3,300

22. What is the present worth of ₹2,400 due in 4 years at a rate of 5% simple interest per annum?
SSC CHSL 03/07/2024 (Shift-01)
(a) ₹ 2,240 (b) ₹ 2,000
(c) ₹ 2,456 (d) ₹ 2,200

23. Find the amount to be paid at the end if ₹4,500 is invested for a period of 4 years at the rate of 8% simple interest per annum.
SSC CHSL 05/07/2024 (Shift-04)
(a) 6,300 (b) 5,940
(c) 6,080 (d) 5,490

24. A sum at simple interest of 5.5% per annum amounts to ₹7,503 in 4 years. Find the sum (in ₹).
SSC MTS 14/09/2023 (Shift-03)
(a) 5,520 (b) 6,150
(c) 5,350 (d) 5,980

25. Ashima borrowed ₹75,000 from a nationalised bank on 26th March 2022 at the rate of 9% per annum simple interest. If she cleared the account on 7th June of the same year, then what amount did she pay?
SSC CHSL 02/08/2023 (Shift-03)
(a) ₹ 75,000 (b) ₹ 76,350
(c) ₹ 71,350 (d) ₹ 75,350

26. A sum of ₹12,000 amounts to ₹15,000 in 5 years at the rate of simple interest. What is the rate of interest per annum?
SSC CHSL 11/08/2023 (Shift-3)
(a) 8% (b) 7%
(c) 6% (d) 5%

27. In how many years will a sum of ₹2,500 invested at 16% simple interest become ₹4,100?
SSC CHSL TIER-II 10/01/2024
(a) 3 (b) 5
(c) 4 (d) 6

28. How many years will it take for Rs.8,400 to amount to Rs.11,928 at a simple interest rate of 7% per annum?
SSC CGL 11/09/2024 (Shift-02)
(a) 8 (b) 5
(c) 2 (d) 6

29. What is the rate of interest per annum for simple interest at which Rs.1,040 amounts to Rs.1,230 in $1\frac{1}{2}$ years?
SSC CGL 17/09/2024 (Shift-02)
(a) $11\frac{7}{39}\%$ (b) $10\frac{7}{39}\%$
(c) $12\frac{7}{39}\%$ (d) $13\frac{7}{39}\%$

30. In how many years will Rs.7,200 amount to Rs.8,928 at a simple interest rate of 8% per annum?
SSC CGL 17/09/2024 (Shift-03)
SSC CHSL 09/07/2024 (Shift-01)
(a) 4 (b) 3
(c) 2 (d) 5

31. A sum of Rs.1,456, when invested for ten years, amounts to Rs.2,366 on maturity. Find the rate of simple interest per annum that was to be paid on the sum invested.
SSC CGL 19/09/2024 (Shift-02)
(a) 5.75% (b) 6.5%
(c) 6.25% (d) 6.15%

32. How long will take for a sum ₹ 450 to accumulate ₹81 in interest at a simple interest of 4.5% per annum?
SSC CHSL 09/08/2023 (Shift-03)
(a) 4.5 years (b) 4 years
(c) 3 years (d) 3.5 years

33. A person borrowed some money on simple interest. After 4 years, he returned $\frac{9}{5}$ of the money to lender. What was the rate of interest?
SC CGL 12/09/2024 (Shift-02)
(a) 20% p.a. (b) 18% p.a.
(c) 17% p.a. (d) 19% p.a.

34. The simple interest on the sum for 12 years is three-fifth of the sum. The rate of interest per annum is:
SSC CHSL 01/07/2024 (Shift-04)
(a) 3% (b) 6%
(c) 5% (d) 4%

35. An amount of Rs.50,000 becomes Rs.72,000 in 4 years on a simple rate of interest. The rate of interest is _____ per annum.
SSC CHSL 03/07/2024 (Shift-03)
(a) 9% (b) 10%
(c) 11% (d) 12%

36. How much time (in years) will it take for an amount of Rs.4,50,000 to yield Rs.45,000 as simple interest at a 5% per annum rate of interest?
SSC CHSL 04/07/2024 (Shift-02)
(a) 5 (b) 3
(c) 2 (d) 4

37. A sum Rs.18,000 gives a simple interest of Rs.3,240 in 2 years and 3 months. The rate of interest per annum is:
SSC CHSL 05/07/2024 (Shift-01)
(a) 12% (b) 7%
(c) 8% (d) 10%

38. If the ratio of principal and simple interest for 5 years is 10 : 7, then the rate of interest per annum is:
SSC CHSL 05/07/2024 (Shift-03)
(a) 10% (b) 15%
(c) 14% (d) 11%

39. Find the simple interest (in ₹) on ₹2000 at 6.5% per annum rate of interest for the period from 14 February 2023 to 28 April 2023
SSC CGL TIER-II (18/01/2025)
(a) 24 (b) 27
(c) 26 (d) 25

40. Find the simple interest (in Rs.) on Rs.2000 at 6.5% per annum rate of interest for the period from 13 February 2023 to 27 April 2023
SSC CGL TIER-II (20/01/2025)
(a) 27 (b) 25
(c) 26 (d) 24

41. In how many years will ₹2,500 be invested at the rate of 12% per annum simple interest, which amounts to ₹4,900?
SSC CPO 28/06/2024 (Shift-01)
(a) 2 (b) 8
(c) 6 (d) 4

42. In what duration (in years) will ₹1500 amount to ₹2250 at simple interest 10% per annum?
SSC Phase XII 25/06/2024 (Shift-04)
(a) 5.5 (b) 5
(c) 3 (d) 3.5

43. A person borrowed some money on simple interest. After 3 years, he returned $\frac{4}{3}$ of the money to the lender. What was the rate of interest?
SSC CHSL TIER-II 02/11/2023
(a) $11\frac{2}{9}\%$ p.a (b) $11\frac{4}{9}\%$ p.a
(c) $11\frac{3}{9}\%$ p.a (d) $11\frac{1}{9}\%$ p.a

44. The ratio of principal and amount is 5: 6 for one year. Find the rate of simple interest per annum?
SSC MTS 04/09/2023 (Shift-03)
(a) 20% (b) 10%
(c) 11% (d) 15%

45. A sum of ₹ 5,345, when invested at a certain rate of simple interest per annum for 10 years, earns an interest of ₹ 2,672.50 on maturity. What is the rate of simple interest per annum?
SSC MTS 05/09/2023 (Shift-01)
(a) 8% (b) 6.5%
(c) 5% (d) 4.5%

46. In how many years will ₹ 2,000 invested at the rate of 11% p.a. simple interest, amount to ₹ 3,100?
SSC MTS 13/09/2023 (Shift-01)
(a) 4 (b) 5
(c) 3 (d) 2

47. How long (in years) will a sum of money invested at 6% per annum simple interest to increase its value by 40%?
SSC CHSL 02/08/2023 (Shift-01)
(a) $6\frac{2}{3}$ (b) $5\frac{2}{3}$
(c) $6\frac{1}{3}$ (d) $4\frac{1}{3}$

48. A man invested ₹25,400 on simple interest for 9 years to obtain a total amount of ₹67,790 on a certain annual rate of interest .What was the rate of interest (rounded off to 2 decimal places) to obtain the above amount ?
SSC CHSL 04/08/2023 (Shift-04)
(a) 14.58% (b) 18.54%
(c) 18.45% (d) 15.84%

49. If the simple interest on ₹2500 is more than the interest on ₹ 1500 by ₹360 in 3 years, then find the rate of interest per annum.
SSC CHSL 08/08/2023 (Shift-04)
(a) 11% (b) 9%
(c) 10% (d) 12%

50. If ₹2,000 is invested at a simple interest rate of 5% per annum and the interest is added to the principal after 10 years. then in how many years will it amount to ₹4,000 at the same rate of interest?
SSC CHSL 14/08/2023 (Shift-01)
(a) $18\frac{2}{3}$ years (b) $16\frac{1}{3}$ years
(c) $16\frac{2}{3}$ years (d) $18\frac{1}{3}$ years

51. If the simple interest at the same interest rate on ₹500 for 4 years and ₹700 for 2 years, combined together, is ₹280, then what is the rate of interest?
SSC CGL 14/07/2023 (Shift-02)
(a) $9\frac{3}{17}\%$ (b) $6\frac{5}{17}\%$
(c) $8\frac{4}{17}\%$ (d) $7\frac{3}{17}\%$

52. If the simple interest for 5 years is equal to 25% of the principal, then the interest will be equal to the principal after _____ years.
SSC CGL 18/07/2023 (Shift-01)
(a) 20 (b) 30
(c) 25 (d) 22

53. In how many years will a sum of ₹9,500 amount to ₹11,780 at the rate of 8% per annum at simple interest?
SSC CGL 18/07/2023 (Shift-03)
(a) 3 (b) 2
(c) 4 (d) 5

54. A person invests ₹9,840 at 5% per annum simple interest to obtain a total amount of ₹12,300. For how many years did he invest the sum?
SSC CGL 20/07/2023 (Shift-03)
(a) 5 years (b) 4.5 years
(c) 3.5 years (d) 3 years

55. A sum of ₹16,875 was lent out at simple interest, and at the end of 1 year 8 months, the total amount was ₹18,000. Find the rate of interest per annum.
SSC Phase XI 27/06/2023 (Shift-04)
(a) 3.7% (b) 4%
(c) 5% (d) 6.66%

56. Simple interest gained on ₹9,000 at the rate of 8% is equal to ₹2,160. Calculate the time.
SSC Phase-XI 30/06/2023 (Shift-04)
(a) 5 years (b) 4 years
(c) 2 years (d) 3 years

57. Sumit lent ₹ 5000 to Ram for 2 years and ₹ 8000 to Mayur for 3 years on simple interest at the same rate of interest and received ₹ 1700 in total from both of them as interest. What is the rate of interest per annum?
SSC MTS 19/05/2023 (Shift-03)
(a) 8% per annum
(b) 4% per annum
(c) 5% per annum
(d) 10% per annum

58. Simple interest on a sum for 5 years is equal to 20 percent of the principal. In how many years interest will be equal to the principal?
SSC CHSL 10/03/2023 (Shift-04)
(a) 12 yrs. (b) 20 yrs.
(c) 25 yrs. (d) 24 yrs.

59. The simple interest on a sum for 8 years is three-fifth of the sum. The rate of interest per annum is:
SSC CGL 23/09/2024 (Shift-02)
(a) $5\frac{1}{2}\%$ (b) $7\frac{1}{2}\%$
(c) 6% (d) 9%

60. The simple interest on a sum for 6 years is $\frac{3}{5}$ part of the sum. The rate of interest per annum is:
SSC CGL 25/09/2024 (Shift-01)
(a) 3% (b) 7%
(c) 10% (d) 5%

61. The simple interest on a sum for 8 years is $\frac{4}{5}$ of the sum. The rate of interest per annum is:
SSC CHSL 03/07/2024 (Shift-04)
(a) 20% (b) 5%
(c) 10% (d) 15%

62. Ram invested a certain sum of money at simple interest. It amounted to $\frac{5}{4}$ of itself in 3 years. What will be its rate percent per annum?
SSC CHSL 08/07/2024 (Shift-04)
(a) 7.25% (b) 8.33%
(c) 9.12% (d) 8.20%

63. A person borrowed some money on simple interest. After 4 years, he returned $\frac{9}{5}$ of the money to the lender. What was the rate of interest?
SSC CGL 21/07/2023 (Shift-02)
(a) 25% p.a. (b) 10% p.a.
(c) 15% p.a. (d) 20% p.a.

64. The simple interest on a sum of ₹ X in 5 years is $\frac{2}{5}$ of the principal. What is the annual rate of interest?
SSC Phase XI 28/06/2023 (Shift-01)
(a) 8% (b) 10%
(c) 5% (d) 12%

65. A sum of money invested at simple interest becomes $\frac{17}{10}$ of itself in 2 years and 6 months. What is the rate of interest per annum?
SSC CGL 05/12/2022 (Shift-03)
(a) 22 % (b) 16 %
(c) 28 % (d) 34 %

66. The simple interest earned on a sum is $\frac{1}{25}$ of the sum, where the number of years of investment is equal to the rate percentage. For how many years was the sum invested?
SSC MTS 12/10/2021 (Shift-02)
(a) 5 (b) 2
(c) 4 (d) 3

67. If the ratio of the principal and the simple interest for 5 years is 10 : 7, then the rate of interest per annum is:
SSC MTS 08/09/2023 (Shift-03)
(a) 16% (b) 14%
(c) 18% (d) 12%

68. A sum of ₹ 10,500 amounts to ₹ 13,825 in $3\frac{4}{5}$ years at a certain rate percent per annum simple interest. What will be the simple interest on the same sum for 5 years at double the earlier rate?
SSC CGL TIER-II 13/09/2019
(a) ₹ 8,470 (b) ₹ 8,750
(c) ₹ 8,670 (d) ₹ 8,560

69. In how much time (in years, rounded off to 2 decimal places) will Rs.10,800 amount to Rs.24,800 if invested at simple interest at the rate of 15.5% per annum?
SSC CGL 24/09/2024 (Shift-01)
(a) 7.72 (b) 8.36
(c) 9.54 (d) 6.89

70. If Rs.1,875 becomes Rs.2,625 in 4 years, what will Rs.24,000 become at the end of 9 years at the same rate of interest, under simple interest?
SSC CGL 26/09/2024 (Shift-01)
(a) Rs.45,600 (b) Rs.21,600
(c) Rs.34,800 (d) Rs.43,200

71. If a sum of Rs.3,260 on simple interest amounts to Rs.5,420 in 6 years, then what will this sum amount to in 4 years at the same rate of interest?
SSC CHSL 09/07/2024 (Shift-02)
(a) Rs.4,700 (b) Rs.3,700
(c) Rs.3,900 (d) Rs.4,500

72. If an amount of ₹80 becomes ₹96 in 2 years in a bank account which gives simple annual interest, then an amount of ₹62,000 will become _____ after 5 years.
SSC CPO 03/10/2023 (Shift-02)
(a) ₹93,000 (b) ₹84,600
(c) ₹82,400 (d) ₹88,200

73. A person borrowed ₹1760 at 6% per annum and ₹2240 at 9% per annum at simple interest for the same period. He had to pay ₹1536 in all as interest. Find the time (in years).
SSC CPO 27/06/2024 (Shift-01)
(a) 3 (b) 4
(c) 5 (d) 2

74. What will be the total due amount on ₹ 25,000 in 2 years if the interest rates are 4% and 5% respectively?
SSC CHSL 04/08/2023 (Shift-01)
(a) ₹ 26,250 (b) ₹ 28,250
(c) ₹ 26,000 (d) ₹ 27,250

75. A invested ₹4400 at 10% and ₹7500 at 8% simple interest. What amount (in ₹) will he get from his investments after 12 years?
SSC CHSL 11/08/2023 (Shift-01)
(a) 24,830 (b) 24,380
(c) 28,380 (d) 28,840

76. At simple interest a sum of ₹ 6400 becomes ₹ 8320 in 3 years. What will ₹ 7200 become in 5 years at the same rate ?
SSC CGL 21/07/2023 (Shift-01)
(a) ₹10200 (b) ₹10600
(c) ₹10800 (d) ₹10400

77. In how many years will the simple interest on a sum of money be equal to the principal at the rate of $2\frac{6}{7}\%$ per annum?
SSC CHSL 01/07/2024 (Shift-01)
(a) 35 years (b) 33 years
(c) 37 years (d) 34 years

78. The difference between the simple interest on a sum of money for $3\frac{1}{2}$ years at the rate of 12% per annum and simple interest on the same sum of money for $4\frac{1}{2}$ years at 10% per annum is ₹360. Find the sum (in ₹).
SSC Phase XII 21/06/2024 (Shift-02)
(a) 16,000 (b) 12,000
(c) 14,000 (d) 10,000

79. A sum of Rs.2,430 gives a simple interest of Rs.1,093.50 in 3 years. The rate of interest per annum is:
SSC CHSL 10/07/2024 (Shift-01)
(a) 15% (b) 10%
(c) 20% (d) 25%

80. The difference between the simple interests on the principal of ₹500 at an interest rate of 5% per annum for 3 years and 4% per annum for 4 years is:
SSC CGL 23/09/2024 (Shift-01)
(a) Rs.7.5 (b) Rs.5
(c) Rs.15 (d) Rs.10

81. The simple interest on a sum of money for $1\frac{1}{2}$ years at 10% per annum is Rs.30 more than the simple interest on the same sum for 1 year at 12% per annum. Find the sum.
SSC CHSL 02/07/2024 (Shift-02)
(a) Rs.1,000 (b) Rs.1,050
(c) Rs.1,250 (d) Rs.1,200

82. An equal sum of money was lent to P and Q at 8.6% per annum for a period of 7 years and 8 years, respectively. If the difference in interest paid by them was Rs.172, then the sum lent to each was:
SSC CHSL 05/07/2024 (Shift-02)
(a) Rs.3,000 (b) Rs.21,15
(c) Rs.1850 (d) Rs.2,000

83. The simple interest on a certain sum for 4 years at 14% per annum is ₹6,160 less than the simple interest on the same sum at the same rate for 8 years. Find the sum.
SSC CPO 28/06/2024 (Shift-03)
(a) ₹ 10,000 (b) ₹ 9,000
(c) ₹ 11,000 (d) ₹ 12,000

84. The simple interest on a certain sum for 4 years at the rate of 15% p.a. is ₹ 4,500 less than the simple interest on the same sum for 6 years at the same rate of interest. Find the sum.
SSC MTS 06/09/2023 (Shift-02)
(a) ₹ 15,000 (b) ₹ 14,000
(c) ₹ 13,000 (d) ₹ 12,000

85. The simple interest on a certain sum at 10% per annum for two years and three years differ by ₹ 500. Find the sum.
SSC MTS 12/09/2023 (Shift-03)
(a) ₹ 6,000 (b) ₹ 4,500
(c) ₹ 5,500 (d) ₹ 5,000

86. The simple interest on a certain sum for 3 years at 8% per annum is ₹9120 less than the simple interest on the same sum for 7 years at 8% per annum. Find the sum.
SSC CHSL 03/08/2023 (Shift-01)
(a) ₹ 28,500 (b) ₹ 28,250
(c) ₹ 28,750 (d) ₹ 28,000

87. The simple interest of a sum of money in 6 years at the rate of 15% per annum is ₹1,440 less than the simple interest accrued on the same sum in 9 years at 18% per annum. What is the sum of money (in ₹)?
SSC CHSL 11/08/2023 (Shift-04)
(a) 1,800 (b) 2,000
(c) 2,700 (d) 2,400

88. What will be the difference between the interest on a sum of ₹2,000 at 7% per annum for 3 years and that on a sum of ₹ 3,200 at 6% per annum for 2 years?
SSC CHSL 20/03/2023 (Shift-01)
(a) ₹ 36 (b) ₹ 44
(c) ₹ 30 (d) ₹ 40

89. The difference of simple interest from two banks on ₹ 8,000 in 3 years is ₹ 800. If the rate of interest per annum in two banks are R_1 and R_2, then what is the value of $R_1 - R_2$? (Where $R_1 > R_2$)
SSC CGL TIER-II 02/03/2023
(a) $5\frac{1}{3}\%$ (b) $3\frac{1}{3}\%$
(c) $1\frac{1}{3}\%$ (d) $2\frac{1}{3}\%$

90. When two equal amounts are deposited for 5 years and 3 years at the rate of 7% and 9% per annum respectively, the difference of their simple interest is ₹ 475. Then find the deposited amount?
SSC CGL 07/03/2020 (Shift-03)
(a) ₹ 5,837.5 (b) ₹ 5,937.5
(c) ₹ 5,992.5 (d) ₹ 6,037.5

91. What is the present worth of ₹ 1,100 due in 2 years at 5% simple interest per annum ?
SSC CGL 26/07/2023 (Shift-03)
(a) ₹ 3,000 (b) ₹ 2,000
(c) ₹ 1,000 (d) ₹ 1,500

92. The simple interest on a sum for 5 years is ₹ 7590. The rate of interest for the first 3 years is 6 percent per annum and for the next 2 years is 14 percent per annum. What is the sum?
SSC MTS 16/06/2023 (SHIFT-03)
(a) ₹ 17000 (b) ₹ 15000
(c) ₹ 14000 (d) ₹ 16500

93. Sum of ₹20000 and ₹40000 are given on simple interest at the rate of 10 percent and 15 percent per annum respectively for three years. What will be the total simple interest?
SSC CGL TIER-II 08/08/2022
(a) ₹36000 (b) ₹32000
(c) ₹28000 (d) ₹24000

94. A borrows a sum of ₹90,000 for 4 years at 5% simple interest. He lends it to B at 7% for 4 years at simple interest. What is his gain (in ₹)?
SSC CGL 21/04/2022 (Shift-03)
(a) 8,000 (b) 7,200
(c) 7,500 (d) 9,000

95. A sum of ₹ 10,500 amounts to ₹ 13,825 in $3\frac{4}{5}$ years at a certain rate percent per annum simple interest. What will be the simple interest on the same sum for 5 years at double the earlier rate?
SSC CGL TIER-II 13/09/2019
(a) ₹ 8,470 (b) ₹ 8,750
(c) ₹ 8,670 (d) ₹ 8,560

96. The simple interest on a sum of money is $\frac{9}{45}$ of the sum. If the number of years is numerically 5 times of rate per cent per annum, then find the rate per cent per annum.
SSC Phase XII 26/06/2024 (Shift-04)
(a) 5% (b) 2%
(c) 7% (d) 3%

97. Archana took a loan of ₹78,000 from a bank on 24th January 2012 at $8\frac{3}{4}\%$ per annum simple interest and paid it back on 18th June 2012. Find the total amount paid by Archana. (Approximated to nearest integer)
SSC CGL 19/07/2023 (Shift-03)
(a) ₹80,723 (b) ₹90,730
(c) ₹85,733 (d) ₹88,730

◆ TYPE – II ◆
CONCEPT OF TIME

98. A certain sum of money becomes 4 times in 12 years when invested at simple interest. In how many years will it become 10 times of itself at the same rate?
SSC MTS 18/10/2021 (Shift-02)
(a) 60 (b) 48
(c) 24 (d) 36

99. A certain sum of money becomes seven times itself when invested at a certain rate of simple interest, in 14 years. How much time (in years, rounded off to 2 decimal places) will it take to become 18 times itself at the same rate?
SSC CGL 12/09/2024 (Shift-01)
(a) 27.33 (b) 39.67
(c) 42.78 (d) 35.5

100. An amount becomes double in 7 years on simple interest. The amount would be four times in ____ years on the same rate of simple interest.
SSC CHSL 01/07/2024 (Shift-02)
(a) 14 (b) 28
(c) 21 (d) 18

101. A sum of money becomes five times its original value in 15 years when invested at a certain simple interest rate. If the sum was invested twice the time at the same rate of interest, what would be the final amount?
SSC CHSL 08/07/2024 (Shift-03)
(a) The money becomes 7 times its original value
(b) The money becomes 9 times its original value
(c) The money becomes 6 times its original value
(d) The money becomes 8 times its original value

102. A person deposited some amount of money in the bank at simple interest. After 20 years, the amount became nine times the amount deposited. In how many years will the amount become 13 times the amount if the interest rate remains the same?
SSC CPO 27/06/2024 (Shift-03)
(a) 30 (b) 25
(c) 28 (d) 23

103. In a certain time, a sum becomes 5 times at the rate of 6% per annum on simple interest. At what rate of interest, the same sum becomes 8 times in the same duration?
SSC MTS 08/09/2023 (Shift-02)
(a) 8.5% (b) 12.5%
(c) 5.5% (d) 10.5%

104. An amount becomes double in 8 years on simple interest. In how many years would ₹25,000 become ₹1,00,000 with the same rate of interest?
SSC CGL 18/07/2023 (Shift-04)
(a) 32 (b) 28
(c) 16 (d) 24

105. A sum of money doubles itself in 4 years at simple interest. In what time will it become 7 times at the same rate of interest?
SSC CHSL TIER-II 26/06/2023
(a) 24 years (b) 23 years
(c) 21 years (d) 25 years

106. A sum of money doubles itself in 7 years at simple interest. In how much time will it becomes 5 times of itself?
SSC CGL 12/12/2022 (Shift-04)
(a) 25 years (b) 28 years
(c) 23 years (d) 21 years

107. At what rate percent per annum simple interest, a sum of money triples itself in 16 years ?
SSC CGL 26/07/2023 (Shift-01)
(a) 11.5% (b) 11%
(c) 12% (d) 12.5%

108. An amount will be tripled itself in 15 years at a certain rate of simple interest per annum. The rate of interest per annum is:
SSC Phase XII 26/06/2024 (Shift-03)
(a) $13\frac{1}{5}\%$ (b) $11\frac{1}{3}\%$
(c) $13\frac{1}{3}\%$ (d) $14\frac{1}{3}\%$

109. A sum of money becomes seven times itself in 16 years at simple interest rate per annum. Find the annual interest rate.
SSC MTS 05/09/2023 (Shift-03)
(a) 39.2% (b) 20.3%
(c) 37.5% (d) 25.8%

110. A financial institute claims that it returns three times the principal in 25 years on a certain rate of simple interest per annum What is the rate of simple interest?
SSC CHSL 03/08/2023 (Shift-04)
(a) 8% (b) 6%
(c) 4% (d) 5%

111. At a certain annual rate (rounded off to the nearest integer) of simple interest, a sum of money becomes nine times its original value in 21 years. By how much does the rate of interest change if the same of money has to becomes 12 times its original value in 25 years?
SSC CHSL 10/08/2023 (Shift-02)
(a) Decreases by 6%
(b) Increases by 6%
(c) Increases by 8%
(d) Decreases by 8%

112. At what annual rate (rounded off to the nearest integer) of simple interest will a sum of money become five times its initial value in 18 years?
SSC CHSL 10/08/2023 (Shift-03)
(a) 18% (b) 20%
(c) 22% (d) 24%

113. The rate at which a sum becomes four times of itself in 12 years at simple interest will be:
SSC CGL 24/07/2023 (Shift-01)
(a) 20% (b) 25%
(b) 35% (d) 30

114. A sum becomes triple in 9 years. Find the annual rate of simple interest.
SSC Phase-XI 30/06/2023 (Shift-03)
(a) $33\frac{1}{3}\%$ (b) $22\frac{2}{9}\%$
(c) $27\frac{1}{9}\%$ (d) $38\frac{2}{9}\%$

115. What is that rate of simple interest at which a sum of money becomes three times of itself in 36 years?
SSC MTS 14/06/2023 (SHIFT-01)
(a) 7.28% (b) 6.86%
(c) 5.55% (d) 6.28%

116. A sum of money becomes $\frac{8}{7}$ of itself in 2 years at a certain rate of simple interest. The rate per annum is:
SSC CGL 20/07/2023 (Shift-01)
(a) $6\frac{3}{7}\%$ (b) $8\frac{1}{7}\%$
(c) $7\frac{1}{7}\%$ (d) $7\frac{3}{7}\%$

117. In a certain duration of time, a sum become 2 times itself at the rate of 5% per annum simple interest. What will be the rate of interest if the same sum becomes 5 times itself in the same duration?
SSC CGL 08/12/2022 (Shift-01)
(a) 20 % (b) 16 %
(c) 10 % (d) 18 %

118. In how many least number of complete years a sum of money become more than four times of itself at the rate of 50 percent per annum on simple interest?
SSC CGL 09/12/2022 (Shift-04)
(a) 9 years (b) 7 years
(c) 6 years (d) 5 years

◆ TYPE – III ◆
RATE IS X% HIGHER OR LOWER

119. An amount of P was put at a certain rate for 4 years. If it had been put at a 6% higher rate for the same period, it would have fetched ₹600 more interest. What value of 2.5 P?
SSC CHSL 10/08/2023 (Shift-04)
(a) ₹3,750 (b) ₹6,250
(c) ₹4,850 (d) ₹2,500

120. A certain sum of money is given at a certain rate of simple interest for 6 years. Had it been given at 7% higher rate, it would have fetched Rs.1,512 more. Find the sum (in ?).
SSC CGL 18/09/2024 (Shift-01)
(a) ₹3,200 (b) ₹3,600
(c) ₹3,100 (d) ₹4,20

121. A certain sum of money is given at a certain rate for 3 years. Had it been given at 5% higher rate, it would have fetched ₹600 more. Find the sum.
SSC CGL 08/12/2022 (Shift-04)
(a) ₹8,000 (b) ₹5,000
(c) ₹6,000 (d) ₹4,000

122. A person lent certain sum of money at the annual rate of 25 percent on simple interest. In 6 years the interest amounted to ₹360 more than the sum lent. What is the sum?
SSC CGL 01/12/2022 (Shift-02)
(a) ₹600 (b) ₹360
(c) ₹720 (d) ₹540

123. If the annual rate of simple interest increases from 11% to $17\frac{1}{2}\%$ then a person's yearly income increases by ₹1,071.20. The simple interest (in ₹) on the same sum at 10% for 5 years is:
SSC CPO 24/11/2020 (Shift-01)
(a) 16,480 (b) 9,120
(c) 7,250 (d) 8,240

124. A man deposited ₹ 10,000 at simple interest in a fund. The interest rate on the fund increases by 4% every year. If the interest rate at the time of the amount deposited was 8%, what is the interest earned (in ₹) by him after 5 years?
SSC CHSL 02/08/2023 (Shift-04)
(a) 6,000 (b) 8,400
(c) 5,800 (d) 8,000

125. A person took a loan at 5% per annum simple interest during the first year and with an increase of 0.5% simple interest every year from the second year onwards. After 4 years, he paid ₹ 4,600 as a total interest to settle the loan completely. How much was the loan?
SSC CGL 17/07/2023 (Shift-03)
(a) 20,000 (b) 19,000
(c) 18,000 (d) 21,000

126. The interest on a certain deposit at 4.5 % p.a, is ₹135 in one year. How much will the additional interest in one year be on the same deposit at 5% p.a?
SSC CGL 24/07/2023 (Shift-03)
(a) ₹ 16 (b) ₹ 15
(c) ₹ 14 (d) ₹ 18

127. A sum of ₹ 800 amounts to ₹ 1,040 in 3 years at simple interest. If the interest rate is increased by 3%, find the amount.
SSC MTS 14/09/2023 (Shift- 02)
(a) ₹ 1112 (b) ₹ 1212
(c) ₹ 1012 (d) ₹ 1102

128. Simple interest on a certain sum is one-fifth of the sum and the interest rate percent per annum is 5 times the number of years. If the rate of interest increases by 3%, then how much will the simple interest (in Rs.) be on Rs.5,600 for 6 years?
SSC CGL 10/09/2024 (Shift-03)
(a) 4668 (b) 4168
(c) 4268 (d) 4368

129. A certain sum of money is given at a certain rate of simple interest for 4 years. Had it been given at 11% higher rate for the same time, it would have fetched ₹1320 more. Find the sum (in ₹).
SSC CPO 28/06/2024 (Shift-02)
(a) 2500 (b) 3000
(c) 4580 (d) 3240

130. A person invested one-fourth of the sum of ₹ 25,000 at a certain rate of simple interest and the rest at 4% p.a. higher rate. If the total interest received for 2 years is ₹ 4,125. What is the rate at which the second sum was invested?
SSC CGL TIER-II 13/09/2019
(a) 9.5% (b) 9.25%
(c) 9.255% (d) 7.5%

◆TYPE – IV◆
MUNNI METHOD

131. A certain sum of money amounts to ₹1,860 in 2 years and to ₹2,130 in $3\frac{1}{2}$ years at simple interest. Find the sum and the rate of interest, respectively.
SSC CHSL 10/07/2024 (Shift-03)
(a) ₹ 1,500, 12%
(b) ₹ 1,200, 10%
(c) ₹ 1,500, 15%
(d) ₹ 1,200, 12%

132. A sum of money at a fixed rate of simple interest amounts to ₹1,630 in 3 years and to ₹1,708 in 4 years. Find the sum (in ₹).
SSC CGL 19/07/2023 (Shift-04)
(a) 1132 (b) 1296
(c) 1396 (d) 1448

133. A sum amounts to ₹7,656 in 4 years and to ₹8,120 in 5 years at a certain simple interest rate per cent per annum. The rate of interest is:
SSC CPO 05/10/2023 (Shift-02)
(a) 4% (b) 3%
(c) 7% (d) 8%

134. A certain sum of money is lent out at simple interest. If that money becomes Rs.24,000 in 2 years and Rs.32,000 in 4 years, the annual rate of interest is:
SSC CHSL 10/07/2024 (Shift-02)
(a) 30% (b) 16%
(c) 25% (d) 20%

135. A certain sum of money amounts to ₹1250 in 4 years and to ₹1520 is $5\frac{1}{2}$ years at a certain rate of simple interest. Find the rate of interest per annum (correct to two places of decimals.)
SSC Phase XII 25/06/2024 (Shift-02)
(a) 33.96% (b) 23.96%
(c) 43.96% (d) 53.96%

136. If a sum of money becomes ₹ 6,000 in 3 years and ₹10,500 in 7 years and 6 months at the same rate of simple interest, then what is the rate of interest?
SSC CHSL 07/08/2023 (Shift-04)
(a) $33\frac{1}{3}\%$ (b) $25\frac{1}{2}\%$
(c) $45\frac{2}{3}\%$ (d) $66\frac{1}{3}\%$

137. A sum of money at simple interest amounts to ₹ 1,200 in 3 years and to ₹ 1,350 in 4 years. The sum is:
SSC CGL 26/07/2023 (Shift-04)
(a) ₹ 650 (b) ₹ 850
(c) ₹ 450 (d) ₹ 750

138. A certain sum, at a certain rate of simple interest, amounts to ₹2,225 in 4 years and to ₹2,606 in 8 years. Find the sum (in ₹).
SSC Phase XI 27/06/2023 (Shift-02)
(a) 1,344 (b) 1,240
(c) 1,143 (d) 1,844

139. What sum of money amounts to ₹2,600 in 2 years and to ₹2,700 in 3 years at a fixed rate of simple interest annually?
SSC Phase XI 30/06/2023 (Shift-02)
(a) ₹2,300 (b) ₹2,200
(c) ₹2,600 (d) ₹2,400

140. On simple interest, a certain sum becomes ₹ 59,200 in 6 years and ₹ 72,000 in 10 years. If the rate of interest had been 2% more, then in how many years would the sum have become ₹76,000?
SSC CGL 12/04/2022 (Shift-01)
(a) 10 (b) 9
(c) 8 (d) 7

141. A certain sum of money lent out at simple interest amounts to ₹ 12,600 in 2 years and ₹ 16,200 in 4 years. The rate percent per annum is:
SSC MTS 11/09/2023 (Shift-03)
(a) 25% (b) 20%
(c) $12\frac{6}{7}\%$ (d) $17\frac{1}{2}\%$

142. A sum of money invested at a certain rate of simple interest per annum amounts to ₹14,522 in seven years and to ₹18,906 in eleven years. Find the sum invested (in ₹).
SSC CGL 17/07/2023 (Shift-02)
(a) 6850 (b) 6900
(c) 6800 (d) 6750

◆TYPE – V◆
BASED ON ALLIGATION

143. A total of Rs.2,00,000 is divided into two parts for investing in different banks on simple interest. One yields 4% p.a. while the other yields 6% p.a. If the total interest at the end of one year is equivalent to 4.7% p.a. on the whole amount, the amount (in Rs.) invested in each bank is _____, respectively.
SSC CGL 18/09/2024 (Shift-02)
(a) 1,30,000 and 70,000
(b) 1,45,000 and 55,000
(c) 1,20,000 and 80,000
(d) 1,60,000 and 40,000

144. Kiran lent ₹4,000 to two persons such that one part of it at the rate of 8% per annum and the remaining part at the rate of 10% per annum. He got an annual simple interest of ₹352. Then the sum lent at 10% is:
SSC CGL 10/09/2024 (Shift-01)
(a) ₹1,800 (b) ₹2,200
(c) ₹2,400 (d) ₹1,600

145. An amount of Rs.8,000 was invested for 2 years, partly in scheme 1 at the rate of 5% simple interest per annum and the rest in scheme 2 at the rate of 4% simple interest per annum. The total interest received at the end was ₹720. The amount of money invested in scheme 1 is:
SSC CHSL 04/07/2024 (Shift-04)
(a) ₹3,640 (b) ₹7,200
(c) ₹4,400 (d) ₹4,000

146. Mr Gogia invested an amount of ₹13,900 divided into two different schemes A and B at the simple interest rate of 14% pa and 11% pa, respectively. If the total amount of simple interest earned in 2 years is ₹3,508, what was the amount invested in scheme B?
SSC Phase XII 21/06/2024 (Shift-01)
(a) ₹7200 (b) ₹7500
(c) ₹6400 (d) ₹6500

147. A borrows ₹20,000 from B and C. B lends him a certain sum of money at 25% simple interest per annum, while C lends him the remaining sum of money at 21% simple interest per annum. The total interest paid by A in one year is ₹4,500. The sum (in ₹) borrowed from B is:

SSC Phase XII 21/06/2024 (Shift-01)

(a) 12,500 (b) 6,000
(c) 9,000 (d) 7,500

148. The simple interest earned on ₹2,800 at a rate of 10% per annum for 3 years is ₹x and ₹1,736 is the simple interest earned on ₹6,200 at a rate of 14% for y years, then the value of $\frac{x}{y}$ is:

SSC CGL TIER-II 26/10/2023

(a) 550 (b) 380
(c) 420 (d) 140

149. A money lender lent ₹30,000 in two parts, one at the rate of 8% and the other at 9%. If the total annual simple interest on the amount is ₹2,650, then what is the amount of money lent the rate of 8%?

SSC CHSL 14/08/2023 (Shift-04)

(a) ₹6,600 (b) ₹5,000
(c) ₹4,500 (d) ₹5,500

150. S deposits a total of ₹ 50,000 in two accounts which give 8% and 12 % simple interest annually, respectively. After one year, he gets a total ₹5,200. How much money does he deposit in the account with a 12 % interest rate?

SSC CGL 25/07/2023 (Shift-04)

(a) 30000 (b) 20000
(c) 32000 (d) 25000

151. A sum of ₹ 3500 is lent out at simple interest in two parts. One part at the annual rate of 7% and another part at- the annual rate of 5%. If the total annual interest is ₹ 210, then what is the sum lent out at the rate of 7 percent?

SSC MTS 20/06/2023 (Shift-02)

(a) ₹1750 (b) ₹2000
(c) ₹1500 (d) ₹2250

152. Rajesh invested ₹10,000 by dividing it into two different investment schemes, A and B, at simple interest rates of 8% and 10%, respectively. If the total interest earned in 2 years is ₹1,680, the amount invested in scheme A is:

SSC CPO 03/10/2023 (Shift-01)

(a) ₹4,000 (b) ₹2,000
(c) ₹6,000 (d) ₹8,000

153. A sum of ₹5,000 was lent to two people, one at the rate of 16% and the other at the rate of 24%. If the simple interest after one year is ₹960, what is the sum lent at the higher rate?

SSC CHSL 11/07/2024 (Shift-02)

(a) ₹1,500 (b) ₹3,000
(c) ₹2,500 (d) ₹2,000

154. A man invests a total sum of ₹10,000 in a company. A part of the sum was invested at 10% simple interest per annum and the remaining part at 15% simple interest per annum. If the total interest accrued to him in two years equals ₹2,400, the sum invested at 15% simple interest per annum is:

SSC CGL 14/07/2023 (Shift-03)

(a) ₹8,000 (b) ₹4,000
(c) ₹6,000 (d) ₹5,000

◆TYPE – VI◆
CONCEPT OF TOTAL SI

155. A person borrowed some money at the rate of 8% per annum for the first two years, at the rate of 11% per annum for the next three years, and at the rate of 16% per annum for the period beyond five years. If he pays a total interest of Rs.21,400 at the end of nine years, how much money did he borrow (round to the nearest unit) ?

SSC CHSL 03/07/2024 (Shift-01)

(a) Rs.11,938 (b) Rs.18,738
(c) Rs.938,15 (d) Rs.18,938

156. Ramesh borrowed some money at rate of 5% per annum for the first four years, 8% per annum for the next six years, and 12% per annum for the period beyond ten years. If the total interest paid by him at the end of twelve years is ₹9,016, then find the money borrowed by Ramesh.

SSC CPO 03/10/2023 (Shift-03)

(a) ₹9,800 (b) ₹9,616
(c) ₹9,816 (d) ₹9,016

157. Ravi borrowed some money at the rate of 5% per anuum for the first three years ,8% per annum for the next two years, and 10% per annum for the period beyond 5 years. If he paid a total simple interest of ₹12,750 at the end of 7 years, then how much money did he borrow?

SSC CHSL 07/08/2023 (Shift-01)

(a) ₹25,000 (b) ₹26000
(c) ₹27000 (d) 24,000

158. A sum of money earns a simple interest at 7.25% per annum for the first eight years, at 8.5% for the next six years, and at 6.5% for the final four years. If the total interest earned during these eighteen years was ₹35,100, what was the original sum invested (in ₹)?

SSC CGL 20/07/2023 (Shift-02)

(a) 25,800 (b) 25,500
(c) 26,400 (d) 26,000

159. The rate of interest for the first 2 years is 6% p.a. for next 3 years is 10% p.a. and for the period beyond 5 years is 12% pa. If a person gets ₹ 12,771 as simple interest after 7 years, then what money did he invest?

SSC CGL TIER-II 16/11/2020)

(a) ₹ 19,450 (b) ₹ 19,350
(c) ₹ 19,300 (d) ₹ 20,000

160. Ramesh has ₹18,000. He deposited ₹7,000 in a bank at the rate of 5% per annum and Rs.6,000 in other bank at the rate of 6% per annum simple interest. If he received ₹1,160 as simple interest at the end of one year, then the rate of interest per annum on rest of the capital is equal to:

SSC CGL 09/09/2024 (Shift-03)

(a) 10% (b) 11%
(c) 9% (d) 8%

161. A man invested a sum of Rs. 5,000 on simple interest for 5 years such that the rate of interest for the first 2 years is10% per annum, for the next 3 years it is 12% per annum How much interest (in Rs), will he earn at the end of 5 years?

SSC CHSL 03/07/2024 (Shift-02)

(a) 2800 (b) 2450
(c) 3000 (d) 3180

162. A person invests ₹ 5,000 at a simple interest rate of 5% per annum for 5 years. At the end of 5 years, she reinvests the entire amount at the same rate of interest for another 3 years. What will be the total interest earned by the person at the end of 8 years?

SSC MTS 13/09/2023 (Shift-03)

(a) ₹ 2,256.50 (b) ₹ 2,156.50
(c) ₹ 2,187.50 (d) ₹ 2,323.50

◆ TYPE – VII ◆
CONCEPT OF EQUAL SI

163. A person invested a total of ₹ 9,000 in three parts at 3%, 4% and 6% per annum on simple interest. At the end of a year, he received equal interest in all three cases. The amount invested at 6% is

SSC CHSL 16/10/2020 (Shift-03)

(a) ₹2,000 (b) ₹3,000
(c) ₹4,000 (d) ₹5,000

164. If the simple interest on Rs.7,200 in 3 years at the rate of 16% per annum equals the simple interest on nnum, give the same interest as ₹9,600 at the rate of x% per annum in 4 years. The value of x is equal to:

SSC CGL 17/09/2024 (Shift-01)

(a) 9 (b) 11
(c) 8 (d) 10

165. In how much time will Rs.5,000 at 4% per annum simple interest produce the same interest as nnum, give the same interest as ₹8,000 in $2\frac{1}{2}$ years at 8% per annum simple interest?

SSC CGL 26/09/2024 (Shift-03)

(a) 8 years (b) 5 years
(c) 4 years (d) 2 years

166. In what time will ₹10,000 at 4% per annum, give the same interest as ₹8,000 does in 4 years at 5% simple interest?

SSC CGL 14/07/2023 (Shift-01)

(a) 5 year (b) 3 year
(c) 4 year (d) 6 year

167. The simple interest on ₹ 800 for 6 years at 5.5% per annum is equal to the simple interest on ₹ 600 at 4% per annum for a certain period of time. The period of time is:
SSC CGL 24/07/2023 (Shift-02)
(a) 11 years (b) 9 years
(c) 12 years (d) 10 years

168. The simple interest earned on ₹7,000 in 2 years at the rate of R% per annum equals to the simple interest earned on ₹5,000 at the rate of 5% per annum in 14 years. The value of R (in percentage) is:
SSC Phase XI 27/06/2023 (Shift-01)
(a) 28% (b) 15%
(c) 25% (d) 20%

169. A man invests an amount of ₹ 1,05,750 at simple interest in the name of his son, daughter and his wife in such a way that they get the same interest after 3, 4 and 5 years, respectively. If the rate of interest is 5% per annum, then the amount invested for the wife is:
SSC CHSL 12/08/2021 (Shift-03)
(a) ₹ 25,000 (b) ₹ 28,000
(c) ₹ 27,000 (d) ₹ 30,000

◆ TYPE – VIII ◆
MISCELLANEOUS QUESTIONS

170. An amount of ₹12,000 was borrowed at a certain rate of simple interest. After four months, ₹ 6,000 more was borrowed and rate of interest on the total principal was doubled than that of the previous rate. At the end of the year, 2800 was paid as the interest. find the rate that was applicable in the initial.
SSC CHSL 12/10/ 2020 (Shift-01)
(a) 14% (b) 16%
(c) 12% (d) 10%

171. A man take a loan of some amount at some rate of simple interest. After three years, the loan amount is doubled and the rate of interest is decreased by 2%. After 5 years, if the total interest paid on the whole is the ₹13,600, which is equal to the same when the first amount was taken for $11\frac{1}{3}$ years, then the loan taken initially is :
SSC CHSL 14/10/2020 (Shift-02)
(a) ₹13,600 (b) ₹12,500
(c) ₹10,000 (d) ₹12,000

172. If a sum of ₹5,000 is taken at a simple rate of interest of 15% per annum for 3 years and another sum of ₹8,000 is taken at a simple interest of 12% per annum for 4 years, the positive difference of the interests paid is:
SSC CGL 09/09/2024 (Shift-02)
(a) ₹1643 (b) ₹1590
(c) ₹1286 (d) ₹1378

173. The interest accrued on a sum of ₹16,000 lent at a simple interest rate of 14% p.a. for the time in those many years, which gives a simple interest of three-fifths of a certain sum at 10% p.a., is:
SSC CGL 11/09/2024 (Shift-01)
(a) ₹13,440 (b) ₹12,970
(c) ₹15,360 (d) ₹14,280

174. S borrowed some amount from R and promised to pay him 8% interest. Then S invested the borrowed amount in a scheme, upon which he earned a profit of 5% after paying R, the principal amount with interest. How much percentage R would have gained, if he would have invested in the scheme directly?
SSC CGL 11/09/2024 (Shift-02)
(a) 13.4% (b) 13%
(c) 14.4% (d) 14%

175. A trader owes a merchant ₹9,810 due in 1 year, but the trader wants to settle the account after 6 months. If the rate of simple interest is 9% per annum, how much cash (in ₹) should he pay?
SSC CGL 13/09/2024 (Shift-02)
(a) 9,550 (b) 9,450
(c) 9,540 (d) 9,405

176. A sum of ₹10,000 is taken as a loan by Rajesh at a rate of 15% p.a. simple interest for 2 years. But Rajesh could not repay it at the agreed time and asked for an extension of two more years. So, the lender included the interest amount for the period as principal for the next two years at the same rate of interest. The total amount paid by Rajesh at the end of 4 years is:
SSC CGL 13/09/2024 (Shift-03)
(a) ₹18,590 (b) ₹16,900
(c) ₹17,650 (d) ₹15,630

177. In a bank, 6% interest is given on the deposited amount. The bank then invests the money in the stock market and is able to earn Rs.30,000 for every Rs.3,00,000. How much money is the bank earning finally if the total amount deposited in the bank is ₹60,00,000?
SSC CGL 17/09/2024 (Shift-03)
(a) Rs.3,00,000 (b) Rs.6,40,000
(c) Rs.2,40,000 (d) Rs.3,60,000

178. Reshma took a loan of Rs.12,00,000 with simple interest for as many years as the rate of interest. If she paid Rs.9,72,000 as interest at the end of the loan period, what was the rate of interest per annum ?
SSC CHSL 01/07/2024 (Shift-02)
(a) 9% (b) 7.5%
(c) 8% (d) 8.7%

179. What will be the amount due on Rs.12,000 in 2 years when the rate of simple interest on successive years is 9%and 10%, respectively?
SSC CHSL 08/07/2024 (Shift-02)
(a) Rs.14,280 (b) Rs.14,350
(c) Rs.14,250 (d) Rs.14,150

180. What will be the amount due on Rs.36,000 in 2 years when the rate of simple interest for successive years is 8%and 6%, respectively?
SSC CHSL 09/07/2024 (Shift-04)
(a) Rs.38,880 (b) Rs.40,320
(c) Rs.41,760 (d) Rs.41,040

181. The simple interest on a certain sum is one-fifth of the sum and the interest rate percent per annum is 3.2 times the number of years. If the rate of interest increases by 4%, then the simple interest (in ₹) on ₹7,500 for 4 years is:
SSC Phase XII 21/06/2024 (Shift-04)
(a) 3,600 (b) 3,000
(c) 2,500 (d) 5,100

182. A sum invested at the rate of 6% simple interest per annum grows to ₹ 1,13,880 in 5 years. What is the simple interest, If the same amount is invested for 8 years at the rate of 9% per annum?
SSC MTS 04/09/2023 (Shift-02)
(a) ₹ 60,565 (b) ₹ 65,392
(c) ₹ 58,050 (d) ₹ 63,072

183. What will be the amount due on ₹ 24,000 in 2 years when the rate of simple interest on successive years is 4% per annum and 5% per annum, respectively?
SSC MTS 06/09/2023 (Shift-03)
(a) ₹ 26,208 (b) ₹ 25,200
(c) ₹ 26,160 (d) ₹ 24,960

184. ₹ 8,000 becomes ₹ 9,800 in three years at simple interest rate. If the rate of Interest become two times of itself, then what will be the amount of the same principal in four years?
SSC MTS 11/09/2023 (Shift-02)
(a) ₹ 13,500 (b) ₹ 11,800
(c) ₹ 12,800 (d) ₹ 10,800

185. Simple interest on a certain sum is one-fourth of the sum and the interest rate per annum is four times the number of years. If the rate of interest increases by 2%, then how much will the simple interest (in ₹) on ₹ 3600, for 6 years.
SSC CHSL 09/08/2023 (Shift-01)
(a) 2562 (b) 2514
(c) 2692 (d) 2592

186. A money-lender claims he lends money at a simple rate of interest of 8% per annum. But he cleverly includes the interest amount in the principal when he calculates it every six months. The effective rate of interest becomes:
SSC CHSL 09/08/2023 (Shift-02)
(a) 8.16% (b) 8.2%
(c) 8.8% (d) 8.25%

187. In 't' years, the simple interest earned on a certain amount at the rate of 10% per annum is $\frac{5}{8}$th of the principle amount if the rate of interest is made three-fourth of 10% and the simple interest remains the same, which of the following statement is true about the changed value of time?
SSC CHSL 14/08/2023 (Shift-03)

(a) It remains the same as 't'.
(b) It becomes three-fourth of the time 't'.
(c) It increases to $8\frac{1}{3}$ years.
(d) It decreases to $5\frac{1}{2}$ years.

188. A man took a loan of ₹32,400 at a certain rate of simple interest per annum. The rate of interest is one-fourth of the number of years for which the loan is taken. If he paid ₹11,664 as interest at the end of the loan period, the rate of interest was:
SSC CGL 19/07/2023 (Shift-02)
(a) 3% (b) 8%
(c) 5% (d) 12%

189. A person borrows ₹20000 at the rate of 6 percent per annum on simple interest for 3 years. He immediately lends it to another person at the rate of 8.5 percent per annum on simple interest for 3 years. What is the value of gain per year?
SSC CHSL 21/03/2023 (Shift-04)
(a) ₹ 1500 (b) ₹ 500
(c) ₹ 800 (d) ₹ 200

190. A person deposited ₹15,600 in a fixed deposit at 10% per annum simple interest. After every second year he adds his interest earned to the principal. The interest at the end of 4 years is:
SSC CGL 12/12/2022 (Shift-03)
(a) ₹6,655 (b) ₹6,864
(c) ₹3,975 (d) ₹3,744

191. The simple interest on a certain sum is $\frac{1}{8}$ of the sum when the number of years is equal to half of the rate percentage per annum. Find the simple interest (in ₹) on ₹15,000 at the same rate of simple interest for 8 years.
SSC CGL 12/04/2022 (Shift-02)
(a) 5800 (b) 5000
(c) 6000 (d) 5250

192. Simple interest on a certain sum is one-fourth of the sum and the interest rate percentage per annum is 4 times the number of years. If the rate of interest increases by 2%, then what will be the simple interest (in ₹) on ₹ 5000 for 3 years?
SSC CGL 13/04/2022 (Shift-01)
(a) 300 (b) 1500
(c) 2000 (d) 1800

193. A salesperson, with a view to promote sales of an item, applies the principal of simple interest. He declares that 300 pieces of the item can be obtained immediately against cash payment, but a customer will get only 200 pieces of the item if he defers the payment for a year. What is the rate percentage of interest on the whole?
SSC MTS 05/10/2021 (Shift-01)
(a) 30% (b) 25%
(c) 50% (d) 40%

194. Two equal sums were lent on simple interest at 6% and 10% per annum respectively. The first sum was recovered two years later than the second sum and the amount in each case was ₹1105. What was the sum (in ₹) lent in each scheme?
SSC CGL 13/08/2021 (Shift-03)
(a) 900 (b) 850
(c) 936 (d) 891

195. A sum amounts to ₹ 14,395.20 at 9.25% p.a. simple interest in 5.4 years. What will be the simple interest on the same sum at 8.6% p.a. in 4.5 years?
SSC CGL TIER-II 12/09/2019
(a) ₹ 3715.20 (b) ₹ 3627
(c) ₹ 3797.76 (d) ₹ 3672

196. Ramu invested some money at 8% simple interest per annum, and it was grown up to Rs.820 in 8 years. If the same sum was invested at 20% simple interest per annum for a period of 10 years, find the amount.
SSC CHSL 10/07/2024 (Shift-04)
(a) Rs.1,000 (b) Rs.1,200
(c) Rs.1,320 (d) Rs.1,500

197. If the simple interest on a certain sum of money for 20 months at the rate of 19.2% per annum exceeds the simple interest on the same sum for 15 months at the rate of 11.2% p.a. by ₹2,920, then what is the sum (in ₹, to the nearest hundreds)?
SSC CPO 29/06/2024 (Shift-02)
(a) 16,620 (b) 12,660
(c) 16,200 (d) 16,660

198. The simple interest on a certain sum at the rate of 12.5% per annum for 6 years is ₹13,500 less than the principal. Find the simple interest.
SSC CGL 25/07/2023 (Shift-01)
(a) ₹13,500 (b) ₹54,000
(c) ₹40,000 (d) ₹40,500

199. The simple interest on an amount for 6 years at 4% p.a. is ₹ 7,500 less than the simple interest on the same amount for 11 years. Find the amount.
SSC CGL 27/07/2023 (Shift-02)
(a) ₹38,000 (b) ₹37,500
(c) ₹37,000 (d) ₹38,500

200. The simple interest on ₹2,460 will be ₹162 less than the simple interest on ₹3,000 at the rate of 5% per annum simple interest. Find the time taken in years.
SSC Phase XI 28/06/2023 (Shift-03)
(a) 4 (b) 6
(c) 5 (d) 3

201. The simple interest on a certain sum for 3 years at 14% p.a. is ₹4,200 less than the simple interest on the same sum for 5 years at the same rate. Find the sum.
SSC CGL 01/12/2022 (Shift-03)
(a) ₹16,000 (b) ₹10,000
(c) ₹15,000 (d) ₹12,000

202. A sum of ₹ 3000 becomes ₹ 6000 when invested in scheme of simple interest. If the annual rate of interest and the number of years for which the sum invested are same, then what is the annual rate of interest?
SSC CHSL 17/03/2023 (Shift-02)
(a) 5% (b) 20%
(c) 10% (d) 15%

203. Kirti invested a sum of money at a certain rate of simple interest per annum for 3 years in "Bachat scheme". Had she invested the same sum in "Shiksha scheme" for 3 years which has a 2% higher simple interest rate per annum, the investment would have fetched ₹540 more. How much did Kirti invest?
SSC MTS 05/07/2022 (Shift-03)
(a) ₹9500 (b) ₹9000
(c) ₹7000 (d) ₹8000

204. A sum was invested at simple interest at x% p.a. for $2\frac{1}{2}$ years. Had it been invested at $(x + 3)$% for the same time, it would have fetched ₹ 585 more. The simple interest on the same sum for $4\frac{2}{3}$ years at 14% p.a. is:
SSC MTS 27/10/2021 (Shift-02)
(a) ₹ 4,732 (b) ₹ 5,096
(c) ₹ 4,900 (d) ₹ 5,460

205. Aman invested a sum of money in scheme A at the rate of 8% p.a. for a period of 5 years on simple interest. He invested four times of the sum of scheme A in scheme B at the rate of 5% p.a. for 7 years on simple interest. He received a total interest amount of Rs.1,620. How much (in Rs.) did he invest in scheme B?
SSC CGL 19/09/2024 (Shift-01)
(a) 2,400 (b) 900
(c) 3,200 (d) 3,600

206. A bank charges simple interest for Rs.K at the rate of $r^{\frac{1}{2}}$ for n^2 years. Another bank charges simple interest for Rs.L at the rate of $r^{\frac{3}{2}}$ for n^3 years and gives the same interest. The ratio between K and L is:
SSC CHSL 02/07/2024 (Shift-01)
(a) $\frac{1}{r} : \frac{1}{n}$ (b) rn:1
(c) $r^2 : n^2$ (d) $a\, r^{\frac{1}{2}} : n^{\frac{1}{2}}$

207. A sum is deposited in a bank which gives simple interest. The sum becomes 1.25 times in 3 years. If there is a requirement of ₹7,60,000 after seven years, how much amount (in ₹) should one deposit to fulfil the requirement?
SSC CGL 07/12/2022 (Shift-02)
(a) 5,20,000 (b) 5,70,000
(c) 4,80,000 (d) 6,00,000

208. A sum of ₹ 5,000 is divided into two parts such that the simple interest on the first part for $4\frac{1}{5}$ years at $6\frac{2}{3}\%$ p.a. is double the simple interest on the second part for $2\frac{3}{4}$ years at 4% p.a. What is the difference between the two parts?
SSC CGL TIER-II 11/09/2019
(a) ₹680 (b) ₹600
(c) ₹560 (d) ₹620

209. A certain amount is lent at x% p.a. simple interest for 3 years. Instead, if the amount was lent at $3x$% p.a. simple interest for 'y' more years, then the simple interest would have been seven times the earlier interest. What is the value of y?
SSC CGL 21/07/2023 (Shift-03)
(a) 3 (b) 4
(c) 5 (d) 6

210. A part of ₹48,500 is invested at a simple interest of 15% per annum. The rest of the money is invested at the rate of 10% simple interest per annum after 2 years of the first investment. The ratio of interest after 5 years from the time when the first amount was invested is 5 : 3. How much is the second part that was invested(in ₹) at the rate of 10% simple interest?
SSC CPO 29/06/2024 (Shift-01)
(a) 20,940 (b) 29,100
(c) 24,900 (d) 19,400

211. What is the ratio of the simple interest earned on a certain amount at the rate of 21% per annum for 8 years to that earned on the same sum at the same rate for 21 years?
SSC CGL TIER-II 03/03/2023
(a) 8 : 21 (b) 21 : 5
(c) 5 : 21 (d) 21 : 8

212. A sum of ₹5,000 is divided into two parts such that the simple interest on the first part for $4\frac{1}{5}$ years at $6\frac{2}{3}\%$ p.a. is double the simple interest on the second part for $2\frac{3}{4}$ years at 4% p.a. The ratio of the second part to the first part is:
SSC CGL TIER-II 03/02/2022
(a) 11 : 14 (b) 11 : 13
(c) 14 : 11 (d) 13 : 11

213. A sum of money was lent in two parts in the ratio 4 : 5 for 4 years and 5 years respectively, both at the rate of 8% per annum simple interest. If the difference between the interests earned from the two parts is ₹ 4680, then what was the total sum lent (in ₹)?
SSC CGL 18/08/2021 (Shift-03)
(a) 58,500 (b) 46,800
(c) 65,000 (d) 42,120

214. A sum of ₹4,000 becomes ₹ 4,960 in 4 years at simple interest. If the rate of interest becomes 1.5 times of itself, then how much this sum will become after 4 years?
SSC CHSL 20/03/2023 (Shift-04)
(a) ₹ 5,440 (b) ₹ 5,680
(c) ₹ 5,280 (d) ₹ 4,890

215. A sum of money becomes three times of itself at the rate of 10 percent per annum on simple interest. Which of the following statement (s) is/are correct?
I. It will become 5 times of itself at the rate of 20 percent per annum on simple interest in the same time period.
II. It will become 7 times of itself at the rate of 30 percent per annum on simple interest in the same time period.
SSC CHSL 21/03/2023 (Shift-01)
(a) Only II
(b) Neither I nor II
(c) Only I
(d) Both I and II

216. ₹ 8,500 becomes ₹ 11,050 in 6 years at a certain rate of simple interest. If the rate becomes 1.8 times of itself, the amount of the same principal in 5 years will be:
SSC CGL 25/07/2023 (Shift-02)
(a) 15,625 (b) 14,350
(c) 13,550 (d) 12,325

217. A man borrowed ₹29,000 for 3 years and ₹45,000 for 5 years at the same rate of simple interest. If he paid a total interest of ₹62,400, then what was the rate of interest?
SSC CPO 29/06/2024 (Shift-03)
(a) 27% (b) 18%
(c) 25% (d) 20%

218. Kanchan lent ₹10,000 for 2 years and ₹5,000 for 4 years at the same rate of simple interest. If the total interest received by Kanchan is ₹2,000, then what was the rate of interest?
SSC CPO 04/10/2023 (Shift-03)
(a) 4% (b) 2%
(c) 3% (d) 5%

219. A person deposited ₹500 for 3 years, ₹ 650 for 5 years, at the same rate of interest and ₹ 1,250 for 7 years. He received a total simple interest of ₹1,620. The rate of interest per annum is:
SSC CGL 17/07/2023 (Shift-01)
(a) 12% (b) 13%
(c) 10.8% (d) 11%

220. The simple interest on a sum of money is four times the principal. The rate percent per annum and time in years are equal. Find the rate percent per annum.
SSC MTS 12/09/2023 (Shift-01)
(a) 15% (b) 20%
(c) 12% (d) 10%

221. Ramu lent Rs.3,000 to Raju and a certain sum to Ravi at the same time and at the same rate of simple interest of 8% per annum. After 5 years, he received a sum of Rs.1,600 from Raju and Ravi together as interest. Find the sum lent to Ravi.
SSC CGL 24/09/2024 (Shift-03)
(a) Rs.1,000 (b) Rs.1,250
(c) Rs.1,200 (d) Rs.1,500

222. A man invested ₹4,500 at 5% per annum and ₹5,500 at 8% per annum simple interest. Both invested were for 4 years each. The amount received from the lower rate of interest is what per cent (rounded off to the nearest integer) of the amount received from higher rate of interest?
SSC CHSL 10/08/2023 (Shift-01)
(a) 63% (b) 74%
(c) 51% (d) 48%

223. The monthly income of Mr. Roy is ₹18,000. He took a loan of ₹30,000 on simple interest for 3 years at the rate of 5% per annum. The amount that he will be paying as simple interest in 3 years is what percent of his monthly salary?
SSC CGL 13/12/2022 (Shift-01)
(a) 30% (b) 35%
(c) 20% (d) 25%

224. $\frac{1}{2}$ of a certain sum is lent at 20 percent rate of interest $\frac{1}{5}$ of the sum is lent at 25 percent rate of interest. The remaining sum is lent at 10 percent rate of interest. If the simple interest received on the total sum for 2 years is ₹360, then what is the sum?
SSC CHSL 09/03/2023 (Shift-03)
(a) ₹ 1200 (b) ₹ 800
(c) ₹ 1000 (d) ₹ 1400

225. If the amount obtained by A by investing ₹ 9,100 for 3 years at a rate of 10% p.a. on simple interest is equal to the amount obtained by B by investing a certain sum of money for five years at a rate of 8% p.a. on simple interest, then 90% of the sum invested by B (in ₹) is:
SSC CGL TIER-II 29/01/2022
(a) 7,800 (b) 7,605
(c) 8,540 (d) 8,450

◆ TYPE – IX ◆
INSTALLMENT

226. What annual installment will discharge a debt of ₹26,160 due in 4 years at 6% simple interest p.a.?
SSC CGL 10/09/2024 (Shift-02)
(a) ₹ 4,500 (b) ₹ 5,500
(c) ₹ 6,000 (d) ₹ 5,800

227. What annual installment (in ₹) will discharge a debt of ₹ 9,429 due in three years at 12.25% simple interest per annum?
Note: Installments will be paid at the end of Year 1, Year 2 and Year 3.
SSC CPO 05/10/2023 (Shift-01)
(a) 2800 (b) 2700
(c) 2840 (d) 2760

228. What is the amount (in ₹) of debt that will be discharged in 6 equal installments of ₹800 each, if the debt is due in 6 years at 5% per annum?
SSC CHSL 08/08/2023 (Shift-01)
(a) 6600 (b) 7500
(c) 8000 (d) 5400

229. Find the total amount of debt that will be discharged by 5 equal instalments of ₹200 each, if the debt is due in 5 years at 5% p.a. at simple interest.
SSC CHSL 02/07/2024 (Shift-03)
(a) ₹ 1400 (b) ₹ 1100
(c) ₹ 1200 (d) ₹ 1255

230. What annual installment will discharge a debt of ₹3,658 due in four years at 12% simple interest per annum?
SSC Phase XII 20/06/2024 (Shift-02)
(a) ₹725 (b) ₹825
(c) ₹750 (d) ₹775

231. Find the amount of equal annual installment, which will discharge a debt of ₹40,376 due in 4 years at the rate of 2% p.a. simple interest.
SSC Phase XII 20/06/2024 (Shift-01)
(a) ₹8,758 (b) ₹10,250
(c) ₹8,600 (d) ₹9,800

232. ₹1,380 has to be paid after 4 years. If four equal installments are required with a simple interest of 10% on each installment, then what is the value of each installment?
SSC Phase XII 20/06/2024 (Shift-04)
(a) ₹ 500 (b) ₹ 200
(c) ₹ 400 (d) ₹ 300

233. A man has to discharge a debt of ₹15,600 which is due in 3 years at 4% simple interest per annum. If he pays this amount in equal installments of annual payment, find the amount for annual payment.
SSC Phase XII 24/06/2024 (Shift-04)
(a) ₹5,100 (b) ₹5,000
(c) ₹5,400 (d) ₹5,200

234. An annual installment of ₹3,500 will discharge a debt of ₹16,310 due in 4 years at y% simple interest per annum. What is the value of y?
[Note: Installments will be paid at the end of Year 1, Year 2, Year 3 and Year 4.
SSC Phase XII 24/06/2024 (Shift-01)
(a) 10.5 (b) 16.5
(c) 11.5 (d) 11

235. To buy a car, Sahil borrowed some amount at 10% simple interest rate. He has paid some loan amount but ₹1,81,500 is due which he has to pay in 3 years. What will be the annual installment (in ₹)?
SSC Phase XI 28/06/2023 (Shift-01)
(a) 65,000 (b) 47,000
(c) 55,000 (d) 54,500

236. What annual installment will discharge a debt of ₹35,455 due in three years at 12% simple interest per annum (Installments will be paid at the end of the years)(Round off to nearest Rupee)?
SSC Phase XI 28/06/2023 (Shift-01)
(a) ₹10,552 (b) ₹14,652
(c) ₹11,804 (d) ₹12,548

237. What is the annual payment Sunil should pay (in ₹) to clear his debt of ₹15,750, which he borrowed from his cousin and promised to return in 3 years at 5% simple interest?
SSC Phase-XI 30/06/2023 (Shift-03)
(a) 4,800 (b) 5,000
(c) 2,500 (d) 4,850

238. A person borrowed ₹2,000 at 5% annual simple interest repayable in 3 equal annual installments. What will be the annual installment?
SSC CGL 13/12/2022 (Shift-03)
(a) $730\frac{10}{63}$ (b) $840\frac{9}{61}$
(c) $640\frac{11}{63}$ (d) $250\frac{10}{63}$

239. A sum of ₹10 is lent by a child to his friend to be returned in 11 monthly installments of ₹1 each, the interest being simple. The rate of interest is:
SSC CGL 13/12/2022 (Shift-02)
(a) $11\frac{9}{11}\%$ (b) $21\frac{9}{11}\%$
(c) $10\frac{2}{11}\%$ (d) $9\frac{1}{11}\%$

240. A car dealer borrows ₹ 20,000 from a bank for a period of 2 years at a simple interest rate of 8% per annum. If he repays the loan in full at the end of the 2-year period, how much will he have to pay?
SSC MTS 04/09/2023 (Shift-01)
(a) ₹ 3,600 (b) ₹ 23,600
(c) ₹ 23,200 (d) ₹ 3,200

241. What five monthly payment will discharge a debt of ₹ 94500 in $1\frac{1}{4}$ year at 12% per annum simple interest?
(a) ₹30,000 (b) ₹31,000
(c) ₹20,000 (d) ₹33,000

242. A computer is available for ₹39,000 on cash payment or ₹19,000 as cash payment followed by five monthly installments of ₹4,200 each. What is the rate of interest per annum under the installment plan?
SSC CHSL 02/08/2023 (Shift-02)
(a) $20\frac{19}{29}\%$ (b) $20\frac{17}{29}\%$
(c) $20\frac{20}{29}\%$ (d) $20\frac{18}{29}\%$

243. Mohan borrows a sum of Rs.4,22,092 at the rate of 20% per annum simple interest. At the end of the first year, he repays Rs.21,679 towards return of principal amount borrowed. If Mohan clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?
SSC CGL 09/09/2024 (Shift-01)
(a) 5,56,367 (b) 5,58,380
(c) 5,61,347 (d) 5,64,914

244. Sohan borrows a sum of Rs.1,41,545 at the rate of 11% per annum simple interest. At the end of the first year, he repays Rs.25,490 towards return of principal amount borrowed. If Sohan clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?
SSC CGL 11/09/2024 (Shift-03)
(a) 1,44,391 (b) 1,36,453
(c) 1,41,222 (d) 1,37,407

245. Pranav borrows a sum of 6,23,305 at the rate of 10% per annum simple interest. At the end of the first year, he repays Rs.16,390 towards return of principal amount borrowed. If Pranav clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in ?) at the end of the second year?
SSC CGL 13/09/2024 (Shift-01)
(a) 7,25,903 (b) 7,29,025
(c) 7,38,700 (d) 7,29,937

246. Aman borrows a sum of Rs.192208 at the rate of 15% per annum simple interest. At the end of the first year, he repays Rs.20196 towards return of principal amount borrowed. If Aman clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?
SSC CGL 18/09/2024 (Shift-03)
(a) 226645 (b) 221777
(c) 221757 (d) 230028

247. Pranav borrows a sum of Rs.598397 at the rate of 20% per annum simple interest. At the end of the first year, he repays Rs.24284 towards return of principal amount borrowed. If Pranav clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?
SSC CGL 19/09/2024 (Shift-03)
(a) 812514 (b) 808615
(c) 818513 (d) 817567

248. A customer pays Rs.975 in instalments. The payment is done each month Rs.5 less than the previous month. If the first instalment is Rs.100, how much time will be taken to pay the entire amount?
SSC CGL 23/09/2024 (Shift-03)
(a) 14 months (b) 15 months
(c) 27 months (d) 26 months

249. Naman borrows a sum of Rs.3,48,020 at the rate of 8% per annum simple interest. At the end of the first year, he repays Rs.15,090 towards return of principal amount borrowed. If Naman clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?
SSC CGL 25/09/2024 (Shift-02)
(a) 3,79,386 (b) 3,87,406
(c) 3,88,931 (d) 3,92,733

250. Pranav borrows a sum of Rs.335180 at the rate of 13% per annum simple interest. At the end of the first year, he repays Rs.23060 towards return of principal amount borrowed. If Pranav clears all pending dues at the end of the second year, including interest payment that accrued during the first year, how much does he pay (in Rs.) at the end of the second year?

SSC CGL 25/09/2024 (Shift-03)

(a) 398876
(b) 393949
(c) 396269
(d) 388076

251. A clock is sold for Rs.550 cash or in the installment scheme, for Rs.250 cash down payment and Rs.310 after one month. Find the rate of interest charged in the installment scheme.

SSC CHSL 04/07/2024 (Shift-01)

(a) 40% (b) 20%
(c) 35% (d) 45%

252. Rahna decided to buy a scooty that costs Rs.40,000. The shopkeeper agreed to sell the scooty under the condition of Rs.25,000 cash down payment of Rs.4,000 each month for four months. Find the rate of interest at which the shopkeeper sold the scooty under the installment scheme.

SSC CHSL 04/07/2024 (Shift-03)

(a) 33% (b) $10\frac{1}{3}\%$
(c) $18\frac{1}{2}\%$ (d) $16\frac{1}{2}\%$

253. A laptop charger is sold for Rs.2,540 in cash or for a down payment of Rs.1,340 in cash together with Rs.1,205 to be paid after one month. Find the rate of interest charged in the instalment scheme.

SSC CHSL 11/07/2024 (Shift-01)

(a) 20% p.a. (b) 10% p.a.
(c) 5% p.a. (d) 15% p.a.

254. In a store, Arun has two options to buy a mobile phone, either he can pay the full amount of ₹2,500 or he may buy at an instalment scheme, i.e. ₹520 in down payment and four equal installments, rate of interest being charged at 25% p.a. simple interest. Arun buys a mobile phone on the installment scheme. What is the monthly installment for Arun?

SSC CPO 05/10/2023 (Shift-03)

(a) 510 (b) 530
(c) 550 (d) 520

255. P borrows a sum of ₹5,00,000 from a bank which charges simple interest at the rate of 18% per annum at the beginning of the year. He pays 11 equal monthly installments of ₹40,000 and mortgages his bike at the end of the year to pay the remaining amount. What is the value of the mortgages bike?

SSC Phase XI 27/06/2023 (Shift-02)

(a) 1,50,000
(b) 1,75,000
(c) 1,25,000
(d) 1,10,000

256. A car with a price of ₹6,50,000 is bought by making some down payment. On the balance, a simple interest of approx mate 10% is charged in lump sum and the money is to be paid in 20 equal annual installments of ₹25,000. How much is the down payment?

SSC CGL 06/12/2022 (Shift-02)

(a) ₹1,55,945 (b) ₹1,95,455
(c) ₹1,94,555 (d) ₹1,45,955

257. A person borrows ₹1,00,000 from a bank at 10% per annum simple interest and clears the debt in five years. If the instalment paid at the end of the first, second, third and fourth years to clear the debt are ₹10,000, ₹20,000, ₹30,000 and ₹40,000, respectively, what amount should be paid at the end of the fifth year to clear the debt?

SSC CGL 12/12/2022 (Shift-01)

(a) ₹38,250 (b) ₹30,000
(c) ₹40,450 (d) ₹36,450

258. A cell phone is available for ₹600 or for ₹300 cash down payment together with ₹360 to be paid after two months. Find the rate of interest charged under this scheme.

SSC CGL 24/09/2024 (Shift-02)

(a) 60% (b) 120%
(c) 20% (d) 50%

259. What annual instalment will discharge a debt of ₹10,192 due in 3 years at 10% simple interest?

SSC CPO 27/06/2024 (Shift-02)

(a) ₹3,144.48 (b) ₹3,044.48
(c) ₹3,088.48 (d) ₹3,188.48

ANSWER KEY

1.	(c)	2.	(b)	3.	(d)	4.	(d)	5.	(a)	6.	(c)	7.	(c)	8.	(d)	9.	(b)	10.	(d)
11.	(a)	12.	(b)	13.	(a)	14.	(d)	15.	(a)	16.	(d)	17.	(c)	18.	(c)	19.	(c)	20.	(c)
21.	(c)	22.	(b)	23.	(b)	24.	(b)	25.	(b)	26.	(d)	27.	(c)	28.	(d)	29.	(c)	30.	(b)
31.	(c)	32.	(b)	33.	(a)	34.	(c)	35.	(c)	36.	(c)	37.	(c)	38.	(c)	39.	(c)	40.	(c)
41.	(b)	42.	(b)	43.	(d)	44.	(a)	45.	(c)	46.	(b)	47.	(a)	48.	(b)	49.	(d)	50.	(c)
51.	(c)	52.	(a)	53.	(a)	54.	(a)	55.	(b)	56.	(d)	57.	(c)	58.	(c)	59.	(b)	60.	(c)
61.	(c)	62.	(b)	63.	(d)	64.	(a)	65.	(c)	66.	(b)	67.	(b)	68.	(b)	69.	(b)	70.	(a)
71.	(a)	72.	(a)	73.	(c)	74.	(d)	75.	(b)	76.	(c)	77.	(a)	78.	(b)	79.	(a)	80.	(b)
81.	(a)	82.	(d)	83.	(c)	84.	(a)	85.	(d)	86.	(a)	87.	(b)	88.	(a)	89.	(b)	90.	(b)
91.	(c)	92.	(d)	93.	(d)	94.	(b)	95.	(b)	96.	(b)	97.	(a)	98.	(d)	99.	(b)	100.	(c)
101.	(b)	102.	(a)	103.	(d)	104.	(d)	105.	(a)	106.	(b)	107.	(d)	108.	(c)	109.	(c)	110.	(a)
111.	(b)	112.	(c)	113.	(b)	114.	(b)	115.	(c)	116.	(c)	117.	(a)	118.	(b)	119.	(b)	120.	(b)
121.	(d)	122.	(c)	123.	(d)	124.	(d)	125.	(a)	126.	(b)	127.	(a)	128.	(d)	129.	(b)	130.	(b)
131.	(a)	132.	(c)	133.	(d)	134.	(c)	135.	(a)	136.	(a)	137.	(d)	138.	(d)	139.	(d)	140.	(b)
141.	(b)	142.	(a)	143.	(a)	144.	(d)	145.	(d)	146.	(c)	147.	(d)	148.	(c)	149.	(b)	150.	(a)
151.	(a)	152.	(d)	153.	(d)	154.	(b)	155.	(d)	156.	(a)	157.	(a)	158.	(d)	159.	(b)	160.	(c)
161.	(a)	162.	(c)	163.	(a)	164.	(a)	165.	(a)	166.	(c)	167.	(a)	168.	(c)	169.	(c)	170.	(d)
171.	(c)	172.	(b)	173.	(a)	174.	(b)	175.	(d)	176.	(b)	177.	(c)	178.	(a)	179.	(a)	180.	(d)

181.	(a)	182.	(d)	183.	(c)	184.	(c)	185.	(d)	186.	(a)	187.	(c)	188.	(a)	189.	(b)	190.	(b)
191.	(c)	192.	(d)	193.	(c)	194.	(b)	195.	(a)	196.	(d)	197.	(c)	198.	(d)	199.	(b)	200.	(b)
201.	(c)	202.	(c)	203.	(b)	204.	(b)	205.	(d)	206.	(b)	207.	(c)	208.	(b)	209.	(b)	210.	(b)
211.	(a)	212.	(c)	213.	(a)	214.	(a)	215.	(d)	216.	(d)	217.	(d)	218.	(d)	219.	(a)	220.	(b)
221.	(a)	222.	(b)	223.	(d)	224.	(c)	225.	(b)	226.	(c)	227.	(a)	228.	(d)	229.	(b)	230.	(d)
231.	(d)	232.	(d)	233.	(b)	234.	(d)	235.	(c)	236.	(a)	237.	(b)	238.	(a)	239.	(b)	240.	(c)
241.	(a)	242.	(c)	243.	(d)	244.	(a)	245.	(d)	246.	(a)	247.	(b)	248.	(b)	249.	(b)	250.	(c)
251.	(a)	252.	(a)	253.	(c)	254.	(d)	255.	(a)	256.	(b)	257.	(b)	258.	(b)	259.	(c)		

SOLUTIONS

1. (c) S.I. $= \frac{P \times R \times T}{100} = \frac{15000 \times 7 \times 3}{100}$
$= 3150$

2. (b) S.I. $= \frac{P \times R \times T}{100} = \frac{5575 \times 8 \times 4}{100}$
$= ₹1784$

3. (d) $SI = \frac{PRT}{100} = \frac{9900 \times 3 \times 9}{100} = ₹2673$

4. (d) Given,
$P = 1250$, $R = 5\frac{1}{2}\%$, $T = 3$ year
$\therefore SI = \frac{P \times R \times T}{100} = \frac{1250 \times 11 \times 3}{2 \times 100}$
$= ₹206.25$

5. (a) $S.I = \frac{P \times R \times T}{100} = \frac{5000 \times 12 \times 14}{100 \times 12}$
$= Rs.700$

6. (c) $\because SI = \frac{P \times R \times T}{100}$
$= \frac{8000 \times 10 \times 6}{100} = ₹4800$

7. (c) $\because SI = \frac{P \times R \times T}{100}$
$\Rightarrow 1800 = \frac{P \times 6 \times 5}{100} \Rightarrow P = ₹\ 6000$
Alternate Method:-
SI = 6% × 5 = 30%
30% = 1800
100% = 6000

8. (d) $\therefore SI = \frac{P \times R \times T}{100}$
$= \frac{5400 \times 4 \times 7}{100} = ₹1512$
Note:- Check the multiple of 9
(54 = 9 × 6)
Only option (d) satisfied.

9. (b) $SI = \frac{P \times R \times T}{100}$
$= \frac{135000 \times 6 \times 27}{100 \times 4} = ₹54675$
Note:- Check the multiple of 9
(27 = 9 × 3)
Only option (b) satisfied.

10. (d) $SI = \frac{P \times R \times T}{100}$
$= \frac{6000 \times 8 \times 21}{100 \times 12} = ₹840$

11. (a) Given,
Principal = Rs.6700, time = 13 months, R = 12%
$SI = \frac{P \times R \times T}{100} = \frac{6700 \times 12 \times 13}{100 \times 12}$
$= 67 \times 13 = 871$

12. (b) Given,
$P = 24000$, $R = 18\%$, $T = \frac{7}{12}$ month
$SI = \frac{P \times R \times T}{100} = \frac{24000 \times 18 \times 7}{100 \times 12}$
$= ₹2520$

13. (a)

May	June	July	August
10	30	31	2

Total days = 73 days
$SI = \frac{12300 \times 12 \times 73}{100 \times 365}$
$= \frac{123 \times 12}{5} = 295.2 \approx ₹300$

14. (d) $\because 10\% = ₹72,780$
$\therefore 100\% = ₹7,27,800$
$\therefore P = ₹7,27,800$

15. (a) Total interest = 15 × 5 = 75%
75% → 69687
100% = 92916

16. (d) SI = 9 × 5 = 45%
45% $\xrightarrow{\times 90}$ 4050
100% → 9000 $\xleftarrow{\times 90}$
$\therefore$ Principal = 9000

17. (c) $16\% = \frac{4}{25}$, $P \times \frac{4}{25} \times 5 = 9600$
$\Rightarrow \frac{4}{5}P = 9600 \Rightarrow P = ₹\ 12000$
Alternate Method:-
SI = 16% × 5 = 80%
80 % $\xrightarrow{\times 120}$ 9600
100 % → 12000 $\xleftarrow{\times 120}$

18. (c) R = 15%, T = 8 years
Then, $P = \frac{SI \times 100}{R \times T} = \frac{23340 \times 100}{15 \times 8}$
$= ₹19450$

19. (c) SI for 6 years = 5% × 6 = 30%
Then,
130% → 4160
1% → 32
100% → 3200.
$\therefore$ Sum = 3200

20. (c) Rate = 9.5% (yearly)
= 19% (for 2 years)
So, $\frac{19}{100} \longrightarrow SI$
$P \longrightarrow P$
A = P + SI = 100 + 19 = 119 units
$\Rightarrow$ 119 units → Rs.21420
$\therefore$ 100 units → Rs.18,000

21. (c) SI = 5 × 4 = 20%
A = P + SI = 120%
120% → 6600
100% → 5500
$\therefore$ Sum → Rs.5500

22. (b) SI = 4 × 5 = 20%
$\Rightarrow$ A = 120% → 2400
$\Rightarrow$ P = 100% → 2000
$\therefore$ Present worth = 2000

23. (b) Let the amount be A
$\because A = P + \frac{P \times R \times T}{100}$
$= 4500 + \frac{4500 \times 8 \times 4}{100}$
$= 4500 + 1440 = 5940$
Alternate Method:
A = 100% + 8 × 4% = 132%
$\therefore A = \frac{4500}{100\%} \times 132\% = 5940$

24. (b) SI = 5.5% × 4 = 22 unit
Amount = 22 + 100 = 122 unit
122 unit = 7503
Sum (100 unit) $= \frac{7503}{122} \times 100 = ₹6150$

25. (b)

March	April	May	June
5	30	31	7

Total time = 73 days
$SI = \frac{P \times R \times T}{100} = 75000 \times \frac{9}{100} \times \frac{73}{365}$
$= ₹1350$
Amount = 75000 + 1350 = ₹76350

26. (d) Given,
Principal = 12000,
Amount = 15000, Time = 5 year
SI = 15000 – 12000 = ₹3000

$$SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 3000 = \frac{12000 \times R \times 5}{100} \Rightarrow R = 5\%$$

27. (c) Total SI = ₹4100 – ₹2500 = ₹1600

$$\therefore SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 1600 = \frac{2500 \times 16 \times T}{100}$$

$\Rightarrow T = 4$ yrs

28. (d) SI = 11928 – 8400 = 3528

$$\text{Time} = \frac{SI \times 100}{P \times R} = \frac{3528 \times 100}{8400 \times 7} = 6 \text{ yrs.}$$

29. (c) SI = (1230 – 1040) = 190

$$\text{Time} = \frac{3}{2} \text{ year}$$

$$\Rightarrow \text{Rate} = \frac{190 \times 100 \times 2}{1040 \times 3} = 12\frac{7}{39}\%$$

30. (b) SI = 8928 – 7200 = Rs.1728

$$SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 1728 = \frac{7200 \times 8 \times T}{100}$$

$$\Rightarrow T = \frac{1728}{72 \times 8} = 3 \text{ yrs.}$$

31. (c) $R = \frac{SI \times 100}{P \times t}$

SI = 2366 – 1456 = 910

$$\therefore R = \frac{910 \times 100}{1456 \times 10} = 6.25\%$$

32. (b) Given,
Principal = 450, S.I. = 81, R = 4.5%

$$SI = \frac{P \times R \times T}{100} \Rightarrow 81 = \frac{450 \times \frac{9}{2} \times T}{100}$$

$$\Rightarrow T = \frac{8100 \times 2}{450 \times 9} = 4 \text{ years}$$

33. (a) $\frac{\text{Amount}}{\text{Principal}} = \frac{9}{5}$, Interest
= 9 – 5 = 4, time = 4 years

$$\therefore SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 4 = \frac{5 \times r \times 4}{100} \Rightarrow r = 20\%$$

34. (c) ATQ,

$$\frac{\text{Simple Interest}}{\text{Principal}} = \frac{3}{5}$$

$$S.I = \frac{P \times R \times T}{100} \Rightarrow 3 = \frac{5 \times r \times 12}{100}$$

$\Rightarrow r = 5\%$

$\therefore$ Rate of interest = 5%

Alternate Method:

S.I $\propto$ r × t

$$\frac{3}{5} = 60\%$$

$$r = \frac{60}{12} = 5\%$$

35. (c) 50000 —(Diff.)— 72000
SI → 22000

$$R = \frac{SI \times 100}{P \times T}$$

$$\text{Rate} = \frac{22000 \times 100}{50000 \times 4} = 11\%$$

36. (c) $SI = \frac{P \times R \times T}{100}$

$$\Rightarrow 45000 = \frac{450000 \times 5 \times T}{100}$$

$\Rightarrow T = 2$ years.

Alternate Method:

SI	:	P
45000	:	450000
1	:	10

$$\therefore T = \frac{SI \times 100}{P \times R} = \frac{1 \times 100}{10 \times 5} = 2 \text{ years}$$

37. (c) $SI = \frac{P \times R \times T}{100}$

$$\Rightarrow 3240 = \frac{18000 \times R \times 27}{100 \times 12}$$

$\Rightarrow R = 8\%$

$\therefore$ Rate of interest = 8%

38. (c) P : SI = 10 : 7

$$\text{S.I for 1 year} = \frac{7}{5} = 1.4$$

$$R\% = \frac{1.4}{10} \times 100 = 14\%$$

39. (c) Given,
Principal = 2000, rate = 6.5%

Time = 14 + 31 + 28 = 73 day

$$\text{Simple interest} = \frac{2000 \times 6.5 \times 73}{100 \times 365}$$

= ₹26

Note:- Here rate is (6.5 = 13 × 0.5) multiple of 13 it means answer is also multiple of 13 only option (c) satisfied.

40. (c) Given,
P = ₹2000, r = 6.5%
t = 15 + 31 + 27 = 73 day

$$SI = \frac{2000 \times 6.5 \times 73}{100 \times 365} = ₹26$$

41. (b) Interest = Amount – principle
= 4900 – 2500 = 2400
ATQ,

$$\Rightarrow 2400 = \frac{2500 \times 12 \times T}{100}$$

$\Rightarrow T = 8$ years

42. (b) Interest = Amount – Principal
= 2250 – 1500 = ₹750

$$T = \frac{750 \times 100}{1500 \times 10} = 5 \text{ years}$$

43. (d) Let, Principal = 3 unit
Amount = 4 unit
SI = 4 – 3 = 1 unit

$$SI = \frac{P \times R \times T}{100} \Rightarrow 1 = \frac{3 \times R \times 3}{100}$$

$$\Rightarrow R = \frac{100}{9} = 11\frac{1}{9}\%$$

Alternate Method:-

$$R = \frac{I}{PT} \times 100 = \frac{1}{3 \times 3} \times 100$$

$$= \frac{100}{9} = 11\frac{1}{9}\%$$

44. (a) Principal : Amount
5 : 6

$$R = \frac{1}{5} \times 100 = 20\%$$

45. (c) $SI = \frac{P \times R \times T}{100}$

$$\Rightarrow 2672.50 = \frac{5345 \times R \times 10}{100}$$

$$\Rightarrow R = \frac{26725}{5345}\% = 5\%$$

46. (b) SI = 3100 – 2000 = ₹ 1100

$$\because SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 1100 = \frac{2000 \times 11 \times T}{100}$$

$\Rightarrow T = 5$ years

47. (a) Value increase by 40% = 5 : 7

$$SI = \frac{P \times R \times T}{100} \Rightarrow 2 = 5 \times \frac{6}{100} \times T$$

$$\Rightarrow T = \frac{20}{3} = 6\frac{2}{3} \text{ years.}$$

Alternate Method:-

S.I $\propto$ r × t

$$\frac{40}{6} = 6\frac{2}{3} \text{yrs.}$$

48. (b) We know that,
Amount = 67790
Principal = 25400
SI = 67790 – 25400 = ₹42390

$$SI = \frac{P \times R \times T}{100}$$

$$\Rightarrow 42390 = \frac{25400 \times R \times 9}{100}$$

$$\Rightarrow R = \frac{42390}{254 \times 9} = 18.54\% \text{ (Approx)}$$

49. (d) Let rate of interest be R%
ATQ,

$$\therefore \frac{2500 \times 3 \times R}{100} = \frac{1500 \times 3 \times R}{100} + 360$$

$\Rightarrow (75 - 45)R = 360 \Rightarrow R = 12\%$

50. (c) SI = 5% × 10 = 50%
100% = 2000
150% = 3000
SI = 4000 – 3000 = 1000
R = 5%

$$\text{ATQ, } 1000 = \frac{3000 \times 5 \times T}{100}$$

$$\Rightarrow T = 6\frac{2}{3} \text{years}$$

$$\text{Total time} = 10 + 6\frac{2}{3} = 16\frac{2}{3} \text{ years}$$

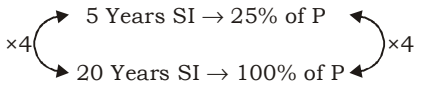
51. (c) We know, $SI = \frac{P \times R \times T}{100}$
ATQ,

$$\frac{500 \times r \times 4}{100} + \frac{700 \times r \times 2}{100} = ₹280$$

$\Rightarrow 34r = ₹280$

$$\Rightarrow r = \frac{280}{34} = 8\frac{4}{17}\%$$

52. (a) ATQ,



In 20 years, interest will be equal to principal.

53. (a) P = ₹9500, Amount = ₹11780
SI = A – P =11780 – 9500 = ₹2280

$$\text{Net SI\%} = \frac{2280}{9500}\times 100\% = 24\%$$

$$\text{Number of years} = \frac{24\%}{8\%} = 3 \text{ year}$$

SMART APPROACH:-
Given,
P = ₹9500 & A = ₹11780
P : A = 25 : 31
Here SI = 6
We know, $8\% = \frac{2 \to \text{SI}}{25 \to \text{P}}$
Required years $= \frac{6}{2} = 3 \text{ year}$

54. (a) SI = 12,300 – 9,840 = 2,460

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$$

$$\Rightarrow 2460 = \frac{9840\times 5\times \text{T}}{100}$$

$$\Rightarrow \text{T} = \frac{24600}{984\times 5} = \frac{100}{4\times 5} = 5 \text{ years}$$

Alternate Method:-
SI = 12,300 – 9,840 = 2,460

$$\text{Net SI \%} = \frac{2460}{9840}\times 100 = 25\%$$

$$\text{Time} = \frac{25\%}{5\%} = 5 \text{ yrs.}$$

55. (b) Given, Amount = ₹18,000
Principal = ₹ 16875
Simple Interest = 18000 – 16875 = 1125
We know that,

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 1125 = \frac{16875\times\text{R}\times 20}{100\times 12}$$

$\Rightarrow$ R = 4%

56. (d) $\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$

$$\Rightarrow 2160 = \frac{9000\times 8\times \text{T}}{100}$$

$\Rightarrow$ Time = 3 years

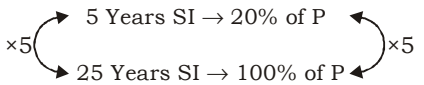
57. (c) ATQ,

$$\Rightarrow \frac{5000\times 2\times \text{R}}{100} + \frac{8000\times 3\times \text{R}}{100} = 1700$$

$\Rightarrow$ 100 R + 240 R = 1700

$$\Rightarrow \text{R} = \frac{1700}{340} = 5\%$$

58. (c)



In 25 yrs. interest will be equal to principal.

59. (b) $\text{SI} = \frac{3}{5}$ Principal $\Rightarrow \frac{\text{SI}}{\text{P}} = \frac{3}{5}$

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 3 = \frac{5\times 8\times \text{r}}{100}$$

$$\Rightarrow \text{r} = 7\frac{1}{2}\%$$

60. (c) Time = 6 year

$\frac{3}{5} \to$ SI
$\frac{5}{5} \to$ Principal

$$\text{S.I of 1 year} = \frac{3}{6} = \frac{1}{2} = 0.5$$

$$\text{R\%} = \frac{0.5}{5}\times 100 = 10\%$$

Alternate Method:
Time = 6 year

$\frac{3}{5} \to$ SI
$\frac{5}{5} \to$ Principal

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$$

$$\Rightarrow 3 = \frac{5\times\text{R}\times 6}{100}$$

$\Rightarrow$ R = 10%

61. (c) $\frac{4}{5} \longrightarrow$ S.I
$\frac{5}{5} \longrightarrow$ Principal

$$\text{S.I for 1years} = \frac{4}{8} = \frac{1}{2} = 0.5$$

$$\text{R\%} = \frac{0.5}{5}\times 100 = 10\%$$

Alternate Method:-

$$\text{S.I} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$$

$$\Rightarrow 4 = \frac{5\times 8\times \text{r}\%}{100}$$

$\therefore$ r% = 10%

62. (b) $\frac{\text{Amount}}{\text{Principal}} = \frac{5}{4}$, time = 3 years.
SI = 5 – 4 = 1

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$$

$$\Rightarrow 1 = \frac{4\times\text{r}\times 3}{100} \Rightarrow \text{r} = 8.33\%$$

$\therefore$ Rate of interest = 8.33%

63. (d) Let, P = ₹ 500 then A = ₹900
SI = 900 – 500 = ₹ 400

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 400 = \frac{500\times\text{R}\times 4}{100}$$

$\Rightarrow$ R = 20%

Alternate Method:-

$$\text{R} = \frac{\text{I}}{\text{PT}}\times 100 = \frac{4}{5\times 4}\times 100 = 20\%$$

64. (a) Given,
Simple interest is $\frac{2}{5}$ of principal
We know that,

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 2 = \frac{5\times\text{R}\times 5}{100}$$

$\Rightarrow$ R = 8%

Alternate Method:-

$$\text{R} = \frac{\text{I}}{\text{PT}}\times 100 = \frac{2}{5\times 5}\times 100 = 8\%$$

65. (c) $\frac{\text{A}}{\text{P}} = \frac{17}{10}$ in $\frac{5}{2}$ yrs.

$$\Rightarrow \frac{\text{I}}{\text{P}} = \frac{7}{10} \text{ in } \frac{5}{2} \text{ yrs.}$$

$$\text{Rate} = \frac{7}{10}\times\frac{2}{5}\times 100 = 28\%$$

SMART APPROACH:-

$$\text{R\%} = \frac{\text{I}}{\text{Pt}}\times 100$$

$$= \frac{7}{2.5\times 10}\times 100$$

$$= \frac{7}{25}\times 100 = 28\%$$

66. (b) Given, simple interest is $\frac{1}{25}$ of sum and no. of years is equal to rate percentage.
We know that,

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 1 = \frac{25\times\text{t}\times\text{t}}{100}$$

$\Rightarrow t^2 = 4 \Rightarrow$ t = 2 years

Alternate Method:-
S.I $\propto$ r × t

$$\frac{1}{25} = 4\% \text{ (r = t)}$$

$t^2 = 4$
t = 2 yrs.

67. (b) P : SI = 10 : 7 = 10x : 7x

$$\therefore \text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100} \Rightarrow 7x = \frac{10x\times\text{R}\times 5}{100}$$

$\Rightarrow$ R = 14%

68. (b) Given,

P_1 = 10,500, R_1 = r%, t = $\frac{19}{5}$ yr & SI_1 = 3325
R_2 = 2r%, t_2 = 5y, SI_2 = ?

$$SI_2 = \frac{3325\times p\times 2r\times 5}{p\times r\times\frac{19}{5}}$$

= 175 × 10 × 5 = ₹8750

69. (b) SI = 24800 – 10800 = 14000

$$\text{SI} = \frac{\text{P}\times\text{R}\times\text{T}}{100}$$

$$\Rightarrow 14000 = \frac{10800\times t\times 31}{100\times 2}$$

$\Rightarrow$ t = 8.36
$\therefore$ Time required = 8.36 years

70. (a) S.I = Amount – Principal
= 2625 – 1875 = 750

$$\text{R} = \frac{750\times 100}{1875\times 4} = 10\%$$

$$\text{A} = 24000 + \frac{24000\times 10\times 9}{100}$$

= 24000 + 21600 = 45600

71. (a) Interest in 6 years = 5420 – 3260 = 2160

$$\text{Interest in 4 years} = \frac{2160}{6}\times 4 = 1440$$

Amount after 4 years = 3260 + 1440 = 4700

72. (a) SI = 96 – 80 = ₹16

$$\therefore \text{R} = \frac{16\times 100}{80\times 2} = 10\%$$

Now,
Eff. Rate = (10% × 5) = 50%

$$\therefore \text{Amount} = \frac{150}{100}\times 62000 = ₹93000$$

73. (c) $SI = \frac{P_1R_1T}{100} + \frac{P_2R_2T}{100}$

$1536 = \frac{1760\times6\times T}{100} + \frac{2240\times9\times T}{100}$

$15360 = 1056\,T + 2016\,T$

$15360 = 3072\,T$

$T = \frac{15360}{3072} = 5$ years

74. (d) Total SI after 2 years = 9% of 25000 = 2250

Total amount payable after 2 years = 25000 + 2250 = ₹27250

Alternate Method:-

SI = 4% + 5% = 9%

100% = 25000

$109\% = \frac{25000}{100}\times 109 = ₹27250$

75. (b) SI after 12 years

$= \frac{4400\times10\times12}{100} + \frac{7500\times8\times12}{100}$

$= 44\times120 + 96\times75$

$= 5280 + 7200 = 12480$

Total Amount = 4400 + 7500 + 12480 = ₹24380

76. (c) Given,

P = 6400, A = 8320

3 Year's SI = 8320 – 6400 = 1920

$\therefore$ 1 year's SI $= \frac{1920}{3} = 640$

Rate % $= \frac{640}{6400}\times 100 = 10\%$

Now,

P = 7200, R = 10%, T = 5

Amount = 7200 × 150% = 10800

77. (a) Rate $= 2\frac{6}{7}\% = \frac{20}{7}\%$

Let S.I = P = 1 unit

ATQ,

$SI = \frac{P\times R\times T}{100} \Rightarrow 1 = \frac{1\times20\times Time}{7\times100}$

$\Rightarrow$ Time = 35 years.

78. (b)

$SI_1 \rightarrow 12\times3\frac{1}{2} = 42\%$

$SI_2 \rightarrow 10\times4\frac{1}{2} = 45\}$ diff. 3%

$3\% \xrightarrow{\times120} 360$

$100\% \xrightarrow{\times120} \boxed{12000}$

79. (a) Principal = 2430

S.I for 3yr. = 1093.50

S.I for 1yr. $= \frac{1093.50}{3} = 364.50$

R% $= \frac{364.50}{2430}\times 100 = 15\%$

Alternate Method:

$S.I = \frac{P\times R\times T}{100}$

$\Rightarrow 1093.50 = \frac{2430\times3\times R}{100}$

$\Rightarrow$ R = 15%

80. (b) Given,

$SI_1 = 3\times5\% = 15\%$

$SI_2 = 4\times4\% = 16\%$

$\Rightarrow$ Difference = 1%

Hence, 1% of 500 = Rs.5

81. (a) SI = RT%

SI for $1\frac{1}{2}$ yrs. $\rightarrow 15\%$

SI for 1 yrs. $\rightarrow 12\%$

$3\% \rightarrow 30$

$1\% \rightarrow 10$

$100\% \rightarrow 1000$

$\therefore$ Sum = Rs.1000

82. (d)

$8.6\% \xrightarrow{\times20} 172$

$100\% \xrightarrow{\times20} \boxed{2000}$

83. (c) Interest $\propto$ Time × Rate

$14(8-4) = 14\times4 = 56$ unit

$56 \text{ unit} \xrightarrow{\times110} 6160$

$100 \text{ unit} \xrightarrow{\times110} \boxed{11000}$

84. (a) Let sum = ₹ P

A.T.Q,

$\frac{P\times6\times15}{100} - \frac{P\times4\times15}{100} = 4500$

$\Rightarrow \frac{P\times30}{100} = 4500 \Rightarrow P = ₹15000$

Alternate Method:-

$SI_1 = 4\times15 = 60\%$

$SI_2 = 6\times15 = 90\}$ − 30%

30% = 4500

100% = 15000

85. (d) 1 year SI = ₹500

$\because\ SI = \frac{P\times R\times T}{100}$

$\Rightarrow 500 = \frac{P\times10\times1}{100} \Rightarrow P = ₹5000$

Alternate Method:-

SI = 10% (3 – 2) = 10%

$10\% \xrightarrow{\times50} 500$

$100\% \xrightarrow{\times50} \boxed{5000}$

86. (a) $SI_1 = 3\times8 = 24\%$

$SI_2 = 7\times8 = 56\}$ − 32%

$32\% \rightarrow ₹\ 9120$

$100\% \rightarrow \frac{9120}{32}\times100 = 28500$

87. (b) $SI_1 = 6\times15 = 90\%$

$SI_2 = 9\times18 = 162\}$ − 72%

$72\% \xrightarrow{\times20} 1440$

$100\% \xrightarrow{\times20} 2000$

88. (a) Difference of SI

$= \frac{2000\times7\times3}{100} - \frac{3200\times6\times2}{100}$

= (420 – 384) = 36

89. (b) $S.I = \frac{P\times R\times T}{100}$

A.T.Q,

$\frac{8000\times3\times R_1}{100} - \frac{8000\times3\times R_2}{100} = 800$

$\Rightarrow \frac{8000\times3}{100}[R_1 - R_2] = 800$

$\Rightarrow R_1 - R_2 = \frac{10}{3} = 3\frac{1}{3}\%$

90. (b) $SI_1 = 5\times7 = 35\%$

$SI_2 = 3\times9 = 27\}$ − 8%

8% = 475

100% = 5937.5

91. (c) In two years –

Total SI = (5 × 2) = 10%

Total due = ₹1100

110% = ₹1100

1% = ₹10

$\therefore$ Present worth = 100% = ₹1000

92. (d) Total SI = 3 × 6 + 2 × 14 = 46%

$46\% \rightarrow 7590$

$100\% \rightarrow \frac{7590}{46}\times100 = 16500$

93. (d) Total SI

$= \frac{20,000\times10\times3}{100} + \frac{40,000\times15\times3}{100}$

= 6000 + 18000 = ₹ 24000

94. (b) $SI_1 = 4\times5 = 20\%$

$SI_2 = 4\times7 = 28\}$ − 8%

Gain = 8% of 90,000 = 7200

95. (b) SI = 13825 –10500 = 3325

$\frac{SI_1}{SI_2} = \frac{P_1R_1T_1}{P_2R_2T_2}$

$\Rightarrow \frac{3325}{SI_2} = \frac{r\times19}{2r\times5\times5}$

$\Rightarrow 175\times50 = SI_2 \Rightarrow SI_2 = 8750$

96. (b) $\frac{I}{P} = \frac{9}{45} = \frac{1}{5} = 20\%$

Rate = x, No. of years = $5x$

$5x\times x = 20$

$\Rightarrow 5x^2 = 20 \Rightarrow x^2 = 4$

$\Rightarrow x = 2\%$

97. (a)

Jan	Feb	March	April	May	June
7	29	31	30	31	18

Total Days = 146 Days

$SI = \frac{78000\times35\times146}{100\times4\times366} = 2722.54$

Amount = 78000 + 2723 (Approx)

= ₹ 80,723

98. (d)

Principal	Amount	SI
1	4	3
1	10	9

3 unit = 12 yrs.

9 unit = 36 yrs.

99. (b)

Principal	Amount	S.I.
1	7	6
1	18	17

6 unit = 14 years
17 unit = 39.67 years

Alternate Method:

Principal and Rate constant
= SI ∝ T

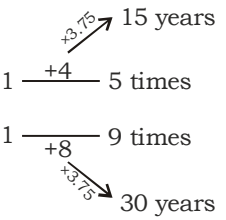
So, $\frac{17}{6} \times 14 = 39.67$

100. (c)

Principal	Amount	S.I.
1	2	1
1	4	3

1 unit = 7 years
3 unit = 21 years

101. (b)



The money becomes 9 times its original value.

102. (a) If amount become 9 times then interest 9 – 1 = 8 time

Principal	Amount	S.I.
1	9	8
1	13	12

8 unit = 20 years
12 unit = 30 years

Alternate Method:

$SI = \frac{P \times R \times T}{100}$

Principal and Rate constant
= SI ∝ T

SI ∝ T
8 — 20
12

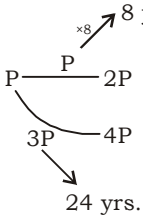
$= \frac{20}{8} \times 12 = 30$

103. (d)

Principal	Amount	S.I.
1	5	4
1	8	7

4 unit = 6%
7 unit = 10.5%

104. (d) 25000 $\xrightarrow{\times 4}$ 100000



105. (a)

Principal	Amount	S.I.
1	2	1
1	7	6

1 unit = 4 years 6 unit = 24 years
Time = 24 yrs.

106. (b)

Principal	Amount	S.I.
1	2	1
1	5	4

1 unit = 7 years
4 unit = 28 years

107. (d) If money will be 3 times then SI will be 2 times
ATQ,

$SI = \frac{P \times R \times T}{100} \Rightarrow 2 = \frac{1 \times R \times 16}{100}$

$\Rightarrow$ Rate = 12.5%

Alternate Method:-

$R = \frac{n-1}{T} \times 100 = \frac{3-1}{16} \times 100 = \frac{25}{2}$

= 12.5%

108. (c) $\because$ $S.I. = \frac{P \times R \times T}{100} \Rightarrow 2 = \frac{1 \times R \times 15}{100}$

$\Rightarrow R = \frac{200}{15} = \frac{40}{3} = 13\frac{1}{3}\%$

109. (c) Let principal = x

$\therefore SI = \frac{P \times R \times T}{100}$

$\Rightarrow 6x = \frac{x \times R \times 16}{100} \Rightarrow R = 37.5\%$

Alternate Method:-

$R = \frac{n-1}{T} \times 100$

$= \frac{7-1}{16} \times 100 = \frac{75}{2} = 37.5\%$

110. (a) $R = \frac{n-1}{T} \times 100$

$= \frac{3-1}{25} \times 100 = 8\%$

111. (b) We know, $R\% = \frac{n-1}{t} \times 100$

ATQ,

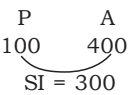
$R_1\% = \frac{8}{21} \times 100\% \approx 38\%$

$R_2 = \frac{11}{25} \times 100\% = 44\%$

% Increase in R% = 44% – 38% = 6%

112. (c) $R\% = \frac{5-1}{18} \times 100 = \frac{200}{9} \approx 22\%$

113. (b)



$SI = \frac{P \times R \times T}{100} \Rightarrow 300 = \frac{100 \times R \times 12}{100}$

$\Rightarrow$ R = 25%

SMART APPROACH:-

$Rate = \frac{(4-1)}{12} \times 100\% = 25\%$

114. (b) P : A = 1 : 3
SI = 3 – 1 = 2

$2 = \frac{1 \times R \times 9}{100} \Rightarrow R\% = \frac{200}{9} = 22\frac{2}{9}\%$

Alternate Method:-

$R\% = \frac{3-1}{9} \times 100 = \frac{200}{9} = 22\frac{2}{9}\%$

115. (c) $R\% = \frac{3-1}{36} \times 100 = \frac{50}{9} = 5.55\%$

$(\frac{1}{9} = 11.11\%)$

116. (c) ATQ,
SI = 8 – 7 = 1
Now,

$1 = \frac{7 \times R \times 2}{100}$

$\Rightarrow R = \frac{50}{7} = 7\frac{1}{7}\%$

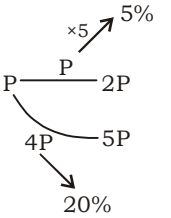
Alternate Method:-

Principal Amount
7 — 8
1

In 1 year = $\frac{1}{7}$

In 2 year = $\frac{1}{14} = 7\frac{1}{7}\%$

117. (a)



Rate = 20%

118. (b) $R\% = \frac{4-1}{T} \times 100 \Rightarrow 50\% = \frac{300}{T}$

$\Rightarrow$ T = 6 yrs.
For more than 4 times time will be greater than 6 yrs.
So, time = 7 yrs.

119. (b) $\frac{P \times (R+6) \times 4}{100} - \frac{P \times R \times 4}{100} = 600$

$\Rightarrow \frac{R+6-R}{100} = \frac{600}{4P} \Rightarrow 6 = \frac{600 \times 25}{P}$

$\Rightarrow$ P = 2500
$\therefore$ 2.5P = 2500 × 2.5 = ₹ 6250

Alternate Method:-

SI = 6 × 4% = 24
24% → 600
100% → 2500
2.5P = 2500 × 2.5 = ₹6250

120. (b) If Simple interest is 7% higher, then simple Interest for 6 year = 7% × 6 = 42%
ATQ,
42% → 1512
1% → 36
100% → 3600
$\therefore$ The principal amount
= Rs. 3600

121. (d) R%, Time = 3 yrs.
R^1 = R + 5%
$\Rightarrow$ For 3 yrs. rate of interest is 15% more.
15% → 600
100% → 4000

122. (c) R = 25%
T = 6 yrs., I = P + 360

ATQ, $P + 360 = \frac{P \times 25 \times 6}{100}$

$\Rightarrow$ 3 P = 2P + 720 $\Rightarrow$ P = 720

Alternate Method:-

$R = 25\% = \frac{1}{4}$

	P	:	A
For one year	4	:	5
For 6 years	24	:	30

$\because$ ATQ,

(30 – 24) unit = (4) unit + 360

(6) unit = (4) unit + 360

(2) unit = 360

Principal (4 unit) = ₹720

SMART APPROACH:-

Let principal amount = 100%

$\therefore$ Total interest = 6 × 25% = 150%

150% – 100% → 360

$\therefore$ 100% → 720

123. (d) Rate difference

$= 17\frac{1}{2}\% - 11\% = 6\frac{1}{2}\%$

$= 1071.2 \times \frac{2}{13} \times 100 = ₹16480$

Now, SI $= \frac{16480 \times 10 \times 5}{100} = ₹\ 8240$

124. (d) Effective rate% = (8 + 12 + 16 + 20 + 24)% = 80%

$\therefore$ Total interest earned in 5 years = 80% of 10,000 = ₹ 8000

125. (a) ATQ, (5 + 5.5 + 6 + 6.5)% = 23%

×200

23 % → 4600

100 % → 20,000

×200

126. (b) ATQ,

×30

4.5 % → 135

5 % → 150

×30

Additional interest = 150 – 135 = 15

127. (a) SI of 3 yrs. = 1040 – 800 = ₹240

$\therefore$ SI $= \frac{P \times R \times T}{100} \Rightarrow 240 = \frac{800 \times R \times 3}{100}$

$\Rightarrow$ R = 10%

Now,

$\Rightarrow R^1$ = (10 + 3) = 13%

SI $= \frac{800 \times 13 \times 3}{100} = 312$

$\therefore$ Amount = 800 + 312 = ₹1112

SMART APPROACH:-

Interest rate is increased by 3% for 3 yrs.

SI = 3 × 3% = 9%

$\Rightarrow$ 9% of 800 = 72

Amount = 1040 + 72 = Rs.1112

128. (d) Let, principal = $5x$, SI = x

Time = t, Rate = 5t

SI $= \frac{P \times R \times t}{100} \Rightarrow x = \frac{5x \times t \times 5t}{100}$

$\Rightarrow$ t = 2 yrs.

Rate = 2 × 5 = 10%

New, rate = (10 + 3)% = 13%

SI $= \frac{5600 \times 13 \times 6}{100}$ = Rs.4368

Note:- To avoid last step calculation, use digital sum method or check the multiple of 7.

56 × 78 = 2 × 6 = 3

Option (d): 4368

digital sum of 4368 = 3.

129. (b) Interest ∝ R × T

11% higher rate means total increment in interest = 11 × 4 = 44%

×30

44% → 1320

100% → 3000

×30

130. (b) Overall interest% $= \frac{4125}{25000 \times 2} \times$

100% $= \frac{33}{4}\%$

r — r + 4

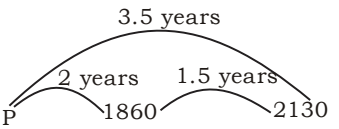
33/4

1 : 3

$\Rightarrow \frac{33}{4} = \frac{r + 3r + 12}{4} \Rightarrow 5.25 = r$

$\therefore$ 2nd Sum was invested at 5.25 + 4 = 9.25%

131. (a)



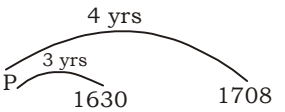
Interest in 1.5 years = 2130 – 1860 = 270

Interest in 2 years = 360

Principal = 1860 – 360 = 1500

Rate $= \frac{180 \times 100}{1500 \times 1} = 12\%$

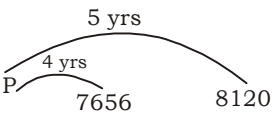
132. (c)



1 Year's SI = 1708 – 1630 = 78

P = 1630 – 78 × 3 = ₹1396

133. (d)



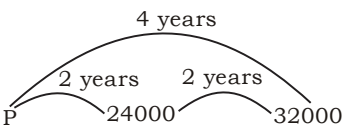
Interest for 1 year = 8120 – 7656 = ₹464

Interest for 4 years = 464 × 4 = 1856

$\therefore$ Principal = 7656 – 1856 = 5800

$\therefore$ R $= \frac{464 \times 100}{5800} = 8\%$

134. (c)

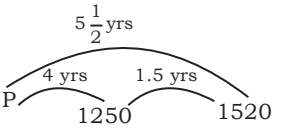


SI in 2 years = 32000 – 24000 = 8000

Principal = 24000 – 8000 = Rs.16000

Rate $= \frac{8000 \times 100}{16000 \times 2} = 25\%$

135. (a)



SI in 1.5 years = 1520 – 1250 = ₹270

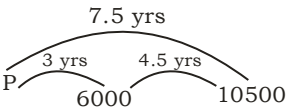
SI in 1 year = ₹ 180

Principal = 1250 – 4 × 180

= 1250 – 720 = ₹ 530

Rate of interest $= \frac{180}{530} \times 100 = 33.96\%$

136. (a)



$\therefore$ SI of 4.5 years = 10,500 – 6000 = 4500

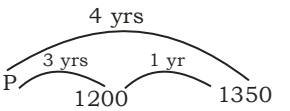
$\therefore$ SI of 3 years = ₹ 3000

$\therefore$ P = 6000 – 3000 = ₹3000

$\therefore$ SI $= \frac{P \times R \times T}{100}$

$\Rightarrow 3000 = \frac{3000 \times R \times 3}{100} \Rightarrow R = 33\frac{1}{3}\%$

137. (d)

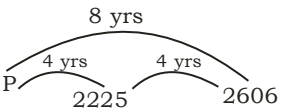


1 Year's SI = ₹1350 – ₹1200 = ₹150

3 Year's SI = ₹150 × 3 = ₹450

$\therefore$ Sum = ₹1200 – ₹450 = ₹750

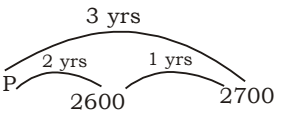
138. (d)



SI in 4 years = ₹381

Principal = 2225 – 381 = ₹1844

139. (d)

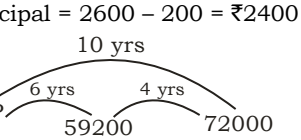


1 years SI = 100

2 years SI = 200

Principal = 2600 – 200 = ₹2400

140. (b)



A.T.Q

Interest of 4 years = 72000 – 59200 = 12800

$\Rightarrow$ Interest of 1 year = 3200

Interest for 6 years = 3200 × 6 = 19200

$\therefore$ P = 59200 – 19200 = ₹40,000

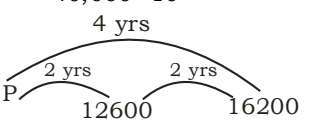
$\therefore$ R% $= \frac{3200 \times 100}{40000} = 8\%$

$\therefore$ Now, (R + 2)% = 10%

Interest = 76,000 – 40,000 = 36,000

Time $= \frac{36,000 \times 100}{40,000 \times 10} = 9$ yrs

141. (b)



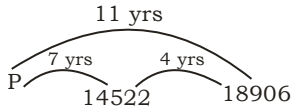
SI of 2 years = 16200 – 12600 = ₹3600

Principal = 12600 – 3600 = ₹ 9000

SI $= \frac{P \times R \times T}{100} \Rightarrow 3600 = \frac{9000 \times R \times 2}{100}$

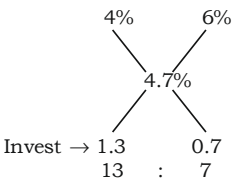
$\Rightarrow$ R = 20%

142. (a)



4 Year's SI = 18906 – 14522 = ₹4384
1 Year's SI = ₹1096
Principal = 14522 – 1096 × 7 = ₹6850

143. (a)



ATQ,
20 units → 2,00,000
13 units → 1,30,000
7 units → 70,000

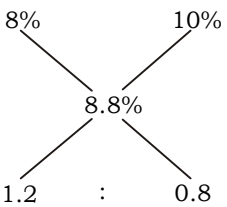
144. (d)

	Ist Part	:	IInd Part
Principal →	x	:	$4000 - x$
Rate →	8%	:	10%

SI = 8% of 4000 = Rs.320
Remaining SI = Rs.352 – Rs.320
= Rs.32 which is due to diff. of rate.
(10 – 8)% = 32
2% = 32
100% = 1600
∴ Sum lent at 10% = Rs.1600

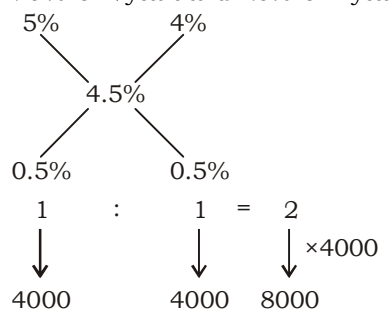
Alternate Method:

Overall rate% = $\frac{352}{4000} \times 100$
= 8.8%



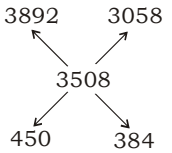
(3 + 2) = 5 unit = 4000
2 unit = 1600

145. (d) Effective rate = $\frac{720}{8000} \times 100 = 9\%$
⇒ 9% for 2 years and 4.5% for 1 year



146. (c) I. In 1st case SI :- $13900 \times \frac{28}{100} = 3892$

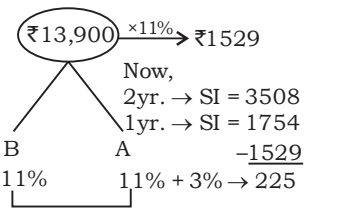
II. In 2nd case SI :- $13900 \times \frac{22}{100} = 3058$



834 unit → 13900
384 unit → 6400

Alternate Method:

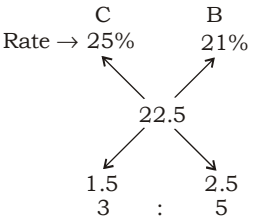
Total amount



$100\% \rightarrow \frac{225}{3} \times 100 = 7500$

Investment made by A = ₹7500
Investment made by B
= 13900 – 7500 = ₹6400

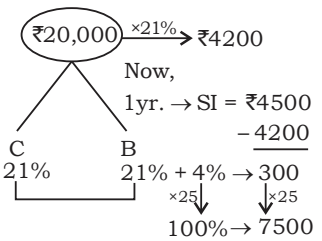
147. (d) Rate = $\frac{4500}{20000} \times 100 = 22.5\%$



(5 + 3) → 8 unit → 20,000
3 unit → 7500

Alternate Method:

Total amount



A borrows money from B = 7500

148. (c) ATQ,

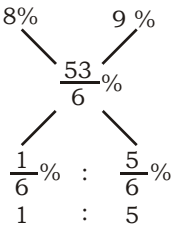
$x = \frac{2800 \times 10 \times 3}{100} = 840$

Again,

$1736 = \frac{6200 \times 14 \times y}{100} \Rightarrow y = 2$

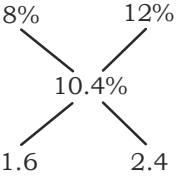
$\therefore \frac{x}{y} = \frac{840}{2} = 420$

149. (b) Net % = $\frac{2650}{30000} \times 100\% = \frac{53}{6}\%$



∵ (1 + 5) unit = 30000
1 unit = 5000
Amount lent on 8% = ₹5000

150. (a) Total SI = $\frac{5200}{50000} \times 100 = 10.4\%$



5 unit → 50000
3 unit → $\frac{50000}{5} \times 3 = ₹30000$

Alternate Method:-

A	:	B
8%	:	8% + 4%

If he deposits whole money at 8%
Interest = $50000 \times \frac{8}{100} = 4000$
Extra = 5200 – 4000 = 1200
4% of B = 1200 ⇒ B = ₹30000

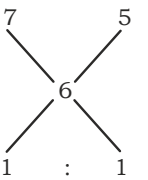
151. (a) ATQ,
Let sum at 7% be ₹ 'A'

$210 = \frac{A \times 7 \times 1}{100} + \frac{(3500 - A) \times 5}{100}$

⇒ 7A – 5A + 17500 = 21000
⇒ 2A = 3500 ⇒ A = ₹1750

Alternate Method:-

Overall Rate% = $\frac{210}{3500} \times 100 = 6\%$



∴ Sum at 7% = $\frac{3500}{2}$ = ₹1750

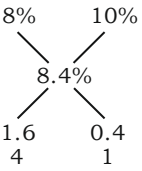
152. (d) SI for 1 year = ₹ 840
If rate = 8%

Then, SI = $\frac{10000 \times 8 \times 1}{100}$ = ₹800

Difference in SI = 840 – 800 = ₹40
This difference in SI is due to difference in rate of interest
(10 – 8)% = ₹ 40
100% = ₹ 2000
Amount invested in scheme A
= 10000 – 2000 = ₹8000

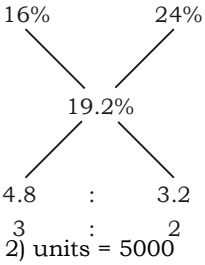
Alternate Method:-

Overall interest% = $\frac{840}{10000} \times 100$
= 8.4%



Amount invested in scheme A
= $\frac{10000}{5} \times 4 = ₹8000$

153. (d) Overall rate% = $\frac{960}{5000} \times 100$
= 19.2%



∵ (3 + 2) units = 5000
∴ 2 unit = 2000

154. (b) Overall interest in one year

$= \frac{1}{2} \times \frac{2400}{10,000} \times 100\% = 12\%$

Sum 1 Sum 2

10% 15%

12%

3 : 2

×2000

5 unit → 10,000

2 unit → 4,000

×2000

155. (d)

9 years

Rate → 8%	11%	16%
Time → 2 years	3 years	4 years
S.I → 16%	33%	64%

Total SI = (16 + 33 + 64)% = 113%

$\therefore P = \frac{21400}{113\%} \times 100\% = 18938$

156. (a)

12 yrs

Time → 4 yrs	6 yrs	2 yrs	
Rate → 5%	8%	12%	
SI → 20%	48%	24%	= 92%

92% = 9016

100% = ₹9800

157. (a)

7 yrs

Time → 3 yrs	2 yrs	2 yrs	
Rate → 5%	8%	10%	
SI → 15%	16%	20%	= 51%

51% = 12750

100% = ₹25000

158. (d)

18 yrs

Time → 8 yrs	6 yrs	4 yrs	
Rate → 7.25%	8.5%	6.5%	
SI → 58%	51%	26%	= 135%

135% = 35100

100% = 26000

159. (b)

7 yrs

Time → 2 yrs	3 yrs	2 yrs	
Rate → 6%	10%	12%	
SI → 12%	30%	24%	= 66%

66% → 12771

$100\% \rightarrow \frac{12771}{66} \times 100 = 19350$ ₹

160. (c)

Sum → 18000

P → 7000	6000	5000	
R →	↓×5%	↓×6%	↓×x%
S.I → 350	+ 360	+ 50x	= 1160

⇒ 50x = 1160 – 710

$\Rightarrow x = \frac{450}{50} = 9\%$

161. (a) ATQ,

$\text{Interest} = \frac{5000 \times 2 \times 10}{100} + \frac{5000 \times 3 \times 12}{100}$

= 1000 + 1800 = 2800

∴ Simple interest for 5 years = 2800

Alternate Method:

Total SI = 2 × 10% + 3 × 12%

= 56%

$\therefore SI = \frac{5000}{100} \times 56 = 2800$

162. (c) $SI = \frac{5000 \times 5 \times 5}{100} = ₹1250$

Amount = 5000 + 1250 = ₹6250

Again,

$SI = \frac{6250 \times 5 \times 3}{100} = ₹937.50$

Total SI = 1250 + 937.50 = ₹2187.50

Alternate Method:-

SI = 5% × 5 = 25%

100% = 5000

125% = 6250

Now,

SI = 5% × 3 = 15%

100% = 6250

115% = 7187.50

SI earned in 8 yrs. = 7187.50 – 5000

= ₹2187.50

163. (a) $SI_1 = SI_2 = SI_3$

$\Rightarrow \frac{a \times 3 \times 1}{100} = \frac{b \times 4 \times 1}{100} = \frac{c \times 6 \times 1}{100}$

⇒ 3a = 4b = 6c

⇒ a : b : c = 4 : 3 : 2

∴ Amount invested at 6%

$= \frac{9000 \times 2}{9} = ₹2000$

SMART APPROACH:-

r → 3	: 4	: 6
p → 24	: 18	: 12
4	: 3	: 2

Amount invested at 6%

$= \frac{9000}{9} \times 2 = \text{Rs.}2000$

Note:- If amount is same then $r \propto \frac{1}{p}$

164. (a) $SI = \frac{P \times R \times T}{100}$

ATQ,

$\frac{7200 \times 3 \times 16}{100} = \frac{9600 \times x \times 4}{100}$

∴ x = 9%

165. (a) ATQ,

$\frac{5000 \times t \times 4}{100} = 8000 \times \frac{5}{2} \times \frac{8}{100}$

⇒ 20t = 20 × 8

⇒ t = 8 years

166. (c) We know,

$SI = \frac{P \times R \times T}{100}$

$= \frac{8000 \times 5 \times 4}{100} = \frac{10000 \times 4 \times t}{100}$

⇒ t = 4 years

167. (a) ATQ,

$\frac{800 \times 6 \times 5.5}{100} = \frac{600 \times 4 \times T}{100}$

⇒ T = 11

168. (c) ATQ,

Simple interest is equal in both case.

$\Rightarrow \frac{7000 \times 2 \times R}{100} = \frac{5000 \times 14 \times 5}{100}$

⇒ R = 25%

169. (c)

	Son		Daughter		Wife
SI ⇒	3x	=	4y	=	5z

$\Rightarrow \frac{x}{20} = \frac{y}{15} = \frac{z}{12}$

Total (47 unit) → 105750

Wife (12 unit) → $\frac{105750}{47} \times 12$

= 27000

170. (d) Total $SI = SI_1 + SI_2$

$2800 = \frac{12000 \times r \times 4}{100 \times 12} + \frac{18000 \times 2r \times 8}{100 \times 12}$

⇒ 2800 = 40r + 240r ⇒ r = 10%

171. (c) $\frac{P \times r \times 3}{100} + \frac{2P \times (r-2) \times 5}{100} = 13,600$ .....(i)

$13,600 = \frac{P \times r \times 34}{100 \times 3}$ .....(ii)

from eqn. (i) and (ii)

r(9 + 30) – 60 = 34r ⇒ r = 12%

Now, $13600 = \frac{P \times 12 \times 34}{100 \times 3}$

⇒ P = ₹10,000

172. (b) Difference

$= \frac{8000 \times 12 \times 4}{100} - \frac{5000 \times 15 \times 3}{100}$

= 3840 – 2250 = Rs.1590

173. (a) Let, certain sum = 500x

$SI = \frac{3}{5} \times x = \frac{3x}{5}$, R = 10%

$T = \frac{SI \times 100}{P \times R} = \frac{3x \times 100}{x \times 5 \times 10} = 6$ yrs.

$SI = \frac{16000 \times 14 \times 6}{100} = \text{Rs.}13440$

174. (b) Let, Principal amount = x

rate of interest = r%

ATQ,

x × r% – x × 5% = x × 8%

⇒ r = 13%

175. (d) Amount = Rs.9810, R = 9%

⇒ for 6 months (R) = 4.5%

ATQ,

109 unit → 9810

$104.5 \text{ unit} \rightarrow \frac{9810}{109} \times 104.5$

$= \frac{981 \times 1045}{109} = 9405$

Used unit digit:- '5'

Only option (d) has unit digit = 5

⇒ (d) Rs.9405

176. (b) Amount after 2 years

$= \frac{10000\times15\times2}{100} + 10000 = 13000$

Now amount becomes principal for next 2 years.

Amount after 4 years

$= 13000 + \frac{13000\times15\times2}{100}$

$= 13000 + 3900 = 16900$

Alternate Method:

P = 10000

SI after 2 years = 15 × 2 = 30%

Now

Amount after 4 years

$= 10000 \times \frac{13}{10} \times \frac{13}{10} = 16900$

177. (c) Rate of Interest for customer = 6%

earning of bank:

$= \frac{30000}{300000} \times 100 = 10\%$

Diff. = 10 – 6 = 4%

$\therefore$ Earnings of bank = 4% of 6000000 $= \frac{4}{100} \times 6000000$

= Rs. 240000

178. (a) $SI = \frac{P\times R\times T}{100}$

$\Rightarrow 972000 = \frac{1200000\times R\times T}{100}$

$\because R = T$

$\Rightarrow R^2 = 81 \Rightarrow R = 9\%$

179. (a) $SI = \frac{P\times R\times T}{100}$

$\therefore SI = \frac{12000\times9\times1}{100} + \frac{12000\times10\times1}{100}$

$= 1080 + 1200 = 2280$

$\therefore$ Amount = 12000 + 2280

= Rs.14280

Alternate Method:

SI = 1 × 9% + 1 × 10% = 19%

$\therefore A = \frac{12000}{100\%} \times 119\% =$ Rs.14280

180. (d) Amount = 36000 +

$\frac{8\times1}{100} \times 36000 + \frac{6\times1}{100} \times 36000$

$= 36000 + 2880 + 2160 = 41040$

Alternate Method:

SI = 8% + 6% = 14%

$\therefore$ Amount $= \frac{36000}{100\%} \times 114\%$

= 41040

181. (a) $\frac{1}{5} = 20\%$

No. of year $\Rightarrow x$, Rate $\Rightarrow 3.2x$

$3.2x^2 = 20\%$

$x^2 = \frac{200}{32} = \frac{25}{4}$ (Time) $x = \frac{5}{2}$

(Rate) $\Rightarrow 3.2x = \frac{5}{2} \times 3.2 = 8\%$

New rate = 8% + 4%↑ = 12%

$SI = \frac{7500\times12\times4}{100} = 75 \times 48 = 3600$

182. (d) Rate of Interest in 5 years

SI = 6 × 5 = 30%

Amount = 130%

Simple interest in 8 years = 8 × 9 = 72%

$\because$ 130% = 113880, 72% = 63072

183. (c) A.T.Q,

$SI = \frac{24000\times1\times4}{100} + \frac{24000\times1\times5}{100}$

$= 960 + 1200 = 2160$

Amount = 24000 + 2160 = ₹ 26160

Alternate Method:-

SI = 4% + 5% = 9%

100% = 24000

$109\% = \frac{24000}{100} \times 109 = 26160$

184. (c) SI = 9800 – 8000 = ₹ 1800

T = 3 years

$\therefore SI = \frac{P\times R\times T}{100}$

$\Rightarrow 1800 = \frac{8000\times R\times3}{100} \Rightarrow R = \frac{15}{2}\%$

ATQ,

$R = \frac{15}{2}\times2 = 15\%$, T = 4 years

$SI = \frac{8000\times15\times4}{100} =$ ₹4800

Amount = 8000 + 4800 = ₹ 12800

Alternate Method:-

3 yrs. interest = 9800 – 8000 – 1800

4 yrs. interest $= \frac{1800}{3} \times 4 = 2400$

At double rate, 4 yrs. interest = 4800

$\therefore$ Amount = 8000 + 4800 = 12800

185. (d) $SI = \frac{P}{4}$, $R = 4T \Rightarrow T = \frac{R}{4}$

$\frac{P}{4} = \frac{P\times R\times R}{100\times4} \Rightarrow R = 10\%$

If, R = 10 + 2 = 12%, T = 6 years,

P = 3600

$SI = \frac{3600\times12\times6}{100} =$ ₹2592

Alternate Method:-

$\therefore$ ₹3600 is multiple of 9 then SI. is also multiple of 9, the (d) option is multiple of 9.

186. (a) R = 8%

$\because$ count in 6 monthly

Then, R = 4%

Effective rate $= 4 + 4 + \frac{4\times4}{100} = 8.16\%$

187. (c) New rate $= \frac{3}{4}\times10\% = \frac{15}{2}\%$

$SI = \frac{5}{8}$th of principal

$\Rightarrow \frac{5}{8} = \frac{1\times\frac{15}{2}\times t}{100} \Rightarrow t = \frac{25}{3} = 8\frac{1}{3}$ years

188. (a) Let, Time = T, $R = \frac{T}{4}$

$\Rightarrow T = 4R$

We know, $SI = \frac{P\times R\times T}{100}$

$\Rightarrow 11664 = \frac{32400\times R\times4R}{100}$

$\Rightarrow R^2 = \frac{11664}{324\times4} = \frac{2916}{324} = 9$

$\Rightarrow R = 3\%$

189. (b) Rate difference = 8.5 – 6 = 2.5%

Profit per year $= \frac{20000\times2.5\times1}{100}$

= ₹500

190. (b) Interest for first 2 yrs. $= 2\times\frac{10}{100} \times$ 15600 = 1560 × 2 = 3120

Principal for 3rd yr. = 15600 + 3120 = 18720

Principal for next 2 yrs. = 18720

$\Rightarrow$ Interest $= 2 \times \frac{18720}{100} \times 10$

$= 1872 \times 2 = 3744$

$\therefore$ Total interest after 4 yrs.

= 3120 + 3744 = 6864

Alternate Method:

Per year rate = 10%

Per 2 years Rate = 20% $= \frac{1}{5}$

	P	:	A
For 2 years →	5	:	6
For 4 years →	25	:	36

×624 ↓ 15600; SI 11 → 6864

191. (c) Given that Interest

$= \frac{1}{8}$ principal

Time $= \frac{\text{Rate}}{2}$ years

$\Rightarrow \frac{1}{8}P = \frac{P\times R}{100} \times \frac{R}{2} \Rightarrow R = 5\%$

Now, Principal = 15000, Time = 8 yrs, Rate = 5%

$S.I = \frac{15000\times8\times5}{100} =$ Rs.6000

192. (d) Let Amount = 1 then simple interest $= \frac{1}{4}$

The rate of interest = R

Then, time $= \frac{R}{4}$ years

Now, $\frac{1}{4} = \frac{\left[1\times R\times\left(\frac{R}{4}\right)\right]}{100}$

$\Rightarrow \frac{1}{4} = \frac{R^2}{400} \Rightarrow R^2 = 100$

$\Rightarrow R = \sqrt{100} = 10\%$

The new rate of interest = 10 + 2 = 12%

Total interest for 3 years = 3 × 12 = 36%

$\therefore SI = \left(\frac{36}{100}\right) \times 5000 = 1800$

SMART APPROACH:-

$R = \frac{I}{Pt} \times 100$

$R = \frac{\frac{1}{4}}{\frac{R}{4}} \times 100\% \Rightarrow R = 10\%$

New R = 10% + 2% = 12%

∴ S.I in 3 years = 36% of 5000 = Rs. 1800

193. (c) Given,
Total = 300 pieces
If he defers payment for a year he get only 200 pieces.

So, $SI = \frac{P \times R \times T}{100}$

$\Rightarrow 100 = \frac{200 \times r \times 1}{100} \Rightarrow r = 50\%$

194. (b) 10% t = 6% (t + 2)
$\Rightarrow 5t = 3(t + 2) \Rightarrow 5t = 3t + 6$
$\Rightarrow t = 3$ years
Now,
130% → 1105

$\therefore 100\% = \frac{1105}{130} \times 100 = 850$

P = ₹850

195. (a) $\frac{14395.2}{(100 + 9.25 \times 5.4)\%} \times (8.6 \times 4.5)\%$

$= \frac{14395.2}{149.95} \times (8.6 \times 4.5) = 3715.20$

196. (d) SI = 8 × 8% = 64%
(100 + 64)% = 820
1% = 5
(100 + 20 × 10)% = 300%
⇒ 5 × 300 = Rs.1500
Amount = Rs.1500

Alternate Method:

SI_1 = 8 × 8% = 64%
SI_2 = 10 × 20% = 200%

$\therefore A_2 = \frac{820}{164\%} \times 300\% = Rs.1500$

197. (c) Interest ∝ Rate × Time
= [(20 × 19.2) – (15 × 11.2)] unit
= 2920
[Both time are in month so do not need to convert it into year divide at last by 12]

$\frac{216}{12}$ unit → 2920
100 unit → 16,200 (Approx)

198. (d) Let P = 100%
SI = RT% = 12.5 × 6 = 75%
ATQ,
100% – 75% = 13500
25% = 13500
SI → 75% = 13,500 ×3 = ₹40,500

199. (b) $\frac{P \times 11 \times 4}{100} - \frac{P \times 6 \times 4}{100} = 7500$

$\Rightarrow \frac{P \times 20}{100} = ₹7500 \Rightarrow P = ₹37500$

Alternate Method:-

SI_1 = 11 × 4% = 44% ↖
− 20%
SI_2 = 6 × 4% = 24% ↙

×5 (20% = 7500 / 100% = 37500) ×5

200. (b) Given,
Difference b/w SI = ₹162
A.T.Q,

$3000 \times \frac{5}{100} \times t - 2460 \times \frac{5}{100} \times t = 162$

$\Rightarrow \frac{5}{100} \times t \times 540 = ₹162$

$\Rightarrow t = 6$ years

201. (c) SI_1 = 3 × 14 = 42% ↖
− 28%
SI_2 = 5 × 14 = 70% ↙
28% = 4200
100% = 15000

202. (c) Sum = 3000 (given)
R% = Time

$SI = \frac{P \times R \times T}{100} \Rightarrow 3000 = \frac{3000 \times R \times R}{100}$

$\Rightarrow R^2 = 100 \Rightarrow R = 10\%$

Alternate Method:-

S.I ∝ r × t
(r = t)
$r^2 = 100\% \Rightarrow r = 10\%$

203. (b) Let, Rate of interest for 3 yrs at Bachat scheme be 'R'
∴ Rate of interest for Siksha scheme R + 2 for 3 yrs
The extra 2% for 3 yrs fetched her ₹540 more
6% = 540
100% = 9000

204. (b) 3% extra rate yields ₹ 585 more S.I

$\Rightarrow (3 \times 2\frac{1}{2})\% \rightarrow 585 \Rightarrow \frac{15}{2}\% \rightarrow 585$

$100\% = \frac{585}{15} \times 2 \times 100 = ₹\ 7800$

$\therefore$ Required SI = $\frac{7800 \times 14}{100} \times \frac{14}{3}$

= ₹5096

205. (d) Let, P be the sum invested in scheme A.

$\Rightarrow \frac{P \times 8 \times 5}{100} + \frac{4P \times 5 \times 7}{100} = 1620$

$\Rightarrow \frac{40}{100}P + \frac{140}{100}P = 1620$

$\Rightarrow 180P = 162000 \Rightarrow P = 900$

∴ Amount invested in B = 4 × 900 = Rs.3600

Alternate Method:-

Let, x be overall SI%

40 35
x
1 : 4

ATQ,

$\Rightarrow \frac{40 - x}{x - 35} = \frac{4}{1} \Rightarrow x = 36\%$

Hence,

$\Rightarrow \frac{36}{100}(P + 4P) = 1620$

$\Rightarrow P = \frac{1620}{36 \times 5} \times 100 = 900$

Hence, 4P = ₹3600

206. (b) ATQ,

$\frac{K \times \sqrt{r} \times n^2}{100} = \frac{L \times r^{3/2} \times n^3}{100}$

$\Rightarrow K \times r^{1/2} = L \times r^{3/2} \times n$

$\Rightarrow \frac{K}{L} = \frac{r^{3/2} \times n}{r^{1/2}} \Rightarrow \frac{K}{L} = \frac{rn}{1} = rn:1$

207. (c) $\frac{Amount}{Principal} = \frac{1.25}{1} = \frac{5}{4}$

$\Rightarrow \frac{I}{P} = \frac{1}{4}$

3 yrs. → $\frac{1}{4}$, 7 yrs. → $\frac{7}{12}$

After 7 yrs, $\frac{Amount}{Principal} = \frac{19}{12}$

$\Rightarrow \frac{76,0000}{x} = \frac{19}{12} \Rightarrow x = ₹\ 4,80,000$

SMART APPROACH:-

$R\% = \frac{n-1}{t} \times 100 = \frac{0.25}{3} \times 100 = \frac{25}{3}\%$

$\therefore 760,000 = P + P \times \frac{25}{3} \times \frac{7}{100}$

$\Rightarrow 760,000 = \frac{19P}{12} \Rightarrow P = Rs.\ 480,000$

208. (b) Let two parts = x, y

$x \times \frac{20}{3} \times \frac{21}{5} = 2 \times y \times 4 \times \frac{11}{4}$

$\Rightarrow 14x = 11y$

$\Rightarrow \frac{x}{y} = \frac{11}{14}$ → 3 unit

×200
25 unit → 5000
3 unit → 600
×200

209. (b) ATQ,

$\frac{P \times x \times 3}{100} \times 7 = \frac{P \times 3x \times (3 + y)}{100}$

$\Rightarrow 7 = 3 + y \Rightarrow y = 4$

210. (b) Let, the first part of the amount = x
Second part of the amount = 48500 – x
ATQ,

$\Rightarrow \frac{x \times 15 \times 5}{(48500 - x) \times 10 \times 3} = \frac{5}{3}$

$\Rightarrow 3x = 2 \times 48500 - 2x$

$\Rightarrow 5x = 2 \times 48500$

$\Rightarrow x = \frac{2 \times 48500}{5} \Rightarrow x = 19400$

The second part = 48500 – 19400 = 29100

211. (a) $S.I = \frac{P \times R \times T}{100}$

ATQ, $\frac{P \times 21 \times 8}{P \times 21 \times 21} = \frac{8}{21} = 8 : 21$

212. (c) $I \times \frac{21}{5} \times \frac{20}{3} = 2 \times II \times \frac{11}{4} \times 4$

$\Rightarrow \frac{II}{I} = \frac{14}{11}$

213. (a) 1st part 4P × 4 × 8% = 128P; 2nd part 5P × 5 × 8% = 200P

72P% → 4680

P = 6500

Total sum lent = 9P

= 6500 × 9 = ₹58500

214. (a) Given,

Sum = 4000

T = 4 years

SI = (4960 – 4000) = 960

$R = \frac{960 \times 100}{4000 \times 4} = 6\%$

SI 960 R 6%

×1.5 → 1440; ×1.5 → 9

Amount = (4000 + 1440) = 5440

215. (d) Amount = 3, SI = 3 – 1 = 2

$2 = \frac{1 \times 10 \times T}{100} \Rightarrow T = 20$ years

(I) $SI = \frac{1 \times 20 \times 20}{100} = 4$, A = 5 times

(II) $SI = \frac{1 \times 30 \times 20}{100} = 6$, A = 7 times

Both statement are correct

Alternate Method:-

$R\% = \frac{3-1}{T} \times 100 \Rightarrow 10\% = \frac{200}{T}$

$\Rightarrow T = 20$ yrs.

(I) $R\% = \frac{5-1}{T} \times 100 \Rightarrow 20\% = \frac{400}{T}$

$\Rightarrow T = 20$ yrs.

(II) $R\% = \frac{7-1}{T} \times 100 \Rightarrow 30\% = \frac{600}{T}$

$\Rightarrow T = 20$ yrs.

Both (I) and (II) are true

216. (d) $SI = \frac{P \times R \times T}{100}$

$\Rightarrow 2550 = \frac{8500 \times R \times 6}{100} \Rightarrow R = 5\%$

New rate = 1.8 × 5% = 9%

Amount = $8500 + \frac{8500 \times 9 \times 5}{100}$

= ₹12325

217. (d) ATQ,

$\frac{29000 \times 3 \times R}{100} + \frac{45000 \times 5 \times R}{100}$

= 62400

870R + 2250R = 62400

3120R = 62400

R = 20%

218. (d) $SI = \frac{P \times R \times T}{100}$

$\Rightarrow 2{,}000 = R\left[\frac{10{,}000 \times 2}{100} + \frac{5000 \times 4}{100}\right]$

$\Rightarrow R[200 + 200] = 2000 \Rightarrow R = 5\%$

219. (a) Let, rate of interest = r%

Total $SI = \frac{500 \times 3 \times r}{100} + \frac{650 \times 5 \times r}{100} + \frac{1250 \times 7 \times r}{100} = 15r + \frac{65}{2}r + \frac{175}{2}r$

= 135r

ATQ, 135r = 1620 $\Rightarrow r = \frac{1620}{135} = 12\%$

220. (b) $\because SI = \frac{P \times R \times T}{100}$

ATQ,

$4 = \frac{1 \times R \times R}{100} \Rightarrow R^2 = 400 \Rightarrow R = 20\%$

221. (a) Total interest for five year = 1600

for 1 year = $\frac{1600}{5}$ = Rs.320

8% = Rs.320

100% = Rs.4000

Raju = Rs.3000

Ravi = 4000 – 3000 = Rs.1000

Alternate method:-

S.I for Raju = $\frac{3000 \times 8 \times 5}{100}$ = 1200

Remaining S.I for Ravi

= 1600 – 1200 = 400

S.I

Ravi → 8 × 5% = 40% → 400

×2.5 → 100%; ×2.5 → 1000

222. (b) Given, Total interest on 4500 = 5 × 4 = 20%

Total interest on 5500 = 8 × 4 = 32%

Amount on lower rate		Amount on higher rate
4500 × 120%	:	5500 × 132%
90	:	121

$\therefore$ Required% = $\frac{90}{121} \times 100 \approx 74\%$

223. (d) 100% = 30000

15% = 4500

Required % = $\frac{4500}{18000} \times 100 = 25\%$

224. (c) L.C.M of 2, 5 = 10

If we assume sum = 10

Principal →	5	2	3
Rate →	20%	25%	10%
SI →	100%	50%	30%

1 yrs $SI = \frac{360}{2} = 180$

$\because$ 180% = 180

Sum (10) = $\frac{180}{180\%} \times 10 = 10 \times 100 = ₹1000$

225. (b) 130% A = 140% of B

$\Rightarrow 13 \times 9100 = 14 \times B \Rightarrow B = 8450$

$\therefore \frac{90}{100} \times B = \frac{90}{100} \times 8450 = ₹7605$

Note:- (90% → 9 × 10%) it means answer will multiple of 9.

Only option (b) satisfied.

226. (c) We know,

$\text{Installment} = \frac{A \times 100}{100t + \frac{rt(t-1)}{2}}$

$= \frac{26160 \times 100}{400 + 24 \times \frac{3}{2}} = \frac{2616000}{436}$

I = Rs.6000

Alternate Method:

Due debt = 26160

Value of each installment = 100 unit

Due debt = 100 + 106 + 112 + 118

= 436 unit = 26160

100 unit = Rs.6000

227. (a) $\text{Installment} = \frac{A \times 100}{100t + \frac{Rt(t-1)}{2}}$

$= \frac{942900}{300 + 36.75\frac{(2)}{2}}$

$= \frac{942900}{336.75} = \frac{1257200}{449} = ₹2800$

Alternate Method:-

Amount = 100% + (100 + 12.25)% + (100 + 12.25 + 12.25)% = 336.75%

336.75% = 9429

100% = ₹2800

228. (d) Let each installment = 100

Amount due = 100 + 105 + 110 + 115 + 120 + 125 = 675

ATQ,

100 → ₹ 800

$\therefore$ 675 → $\frac{800 \times 675}{100} = ₹5400$

229. (b) Total amount of debt = 100% + 105% + 110% + 115% + 120%

= 550%

100% = 200 (given)

550% = 1100

230. (d) Debt = 3658

Let, installment = 100 unit

Then due debt = (100 + 112 + 124 + 136) = 472 unit

Now,

472 unit = 3658

100 unit = $\frac{3658}{472} \times 100 = 775$

231. (d) Due debt = 40376

Value of each installment = 100 unit

Due debt = 100 + 102 + 104 + 106

= 412 unit → 40376

100 unit = 9800

232. (d) Due debt = 1380

Value of each installment = 100 unit

Due debt = 100 + 110 + 120 + 130

= 460 unit = 1380

100 unit = 300

233. (b) Due debt = 15600
Value of each installment = 100 unit
Due debt = 100 + 104 + 108 = 312
= 312 unit = 15600
100 unit = 5000

234. (d) 3500 × 4 = 14000
Interest = (16310 – 14000) = 2310
Let, y = rate of interest
IV → 3y
III → 2y
II → y
I → 0
6y% × 3500 = 2310
y = 11%

235. (c) Due debt = ₹181500
Let the value of each installment = 100 unit
Due debt = 100 + 110 + 120 = 330 unit
Value of each installment
$= \frac{181500}{330} \times 100 = ₹55000$

236. (a) Due debt = ₹35,455
Let the value of each installment = 100 unit
Due debt = 100 + 112 + 124 = 336 unit
Value of each installment
$= \frac{35455}{336} \times 100 = ₹10552$

237. (b) Due debt = ₹15750
Let the value of each Installment = 100 units
Due debt = 100 + 105 + 110 = 315
Value of each Installment
$= \frac{15750}{315} \times 100 = ₹5000$

238. (a) $\text{Installment} = \frac{100 \times A}{100t + \frac{Rt(t-1)}{2}}$

$\text{Amount} = 2000 + \frac{2000 \times 5 \times 3}{100}$
$= 2000 + 300 = 2300$

$\therefore \text{Installment} = \frac{2300 \times 100}{300 + 15}$

$= \frac{230000}{315} = 730\frac{10}{63}$

Alternate Method:-
Amount payable after 3 years
$= 2000 + \frac{2000 \times 5 \times 3}{100} = 2300$
Due debt = 2300
Value of each installment = 100 unit
Due debt = 100 + 105 + 110
$\text{Installment (100 unit)} = \frac{2300}{315} \times 100$
$= ₹\ 730\frac{10}{63}$

239. (b) Let Rate = R%

$1 = \left[\frac{10 \times R \times 1}{12 \times 100} + \frac{9 \times R \times 1}{12 \times 100} + ... + \frac{1 \times R \times 1}{12 \times 100}\right]$

$1 = \frac{R}{12 \times 100}[10 + 9 + 8 + ... + 1]$

$1 = \frac{R}{1200} \times 55 \Rightarrow R = \frac{240}{11} = 21\frac{9}{11}\%$

240. (c) $\because SI = \frac{P \times R \times T}{100}$

$SI = \frac{20000 \times 8 \times 2}{100} = ₹\ 3200$

$\therefore$ Amount = 20000 + 3200 = ₹23200

Alternate Method:-
SI = 2 × 8% = 16%

×200
100 % → 20,000
116 % → 23,200
×200

241. (a) $r = \frac{12}{12} \times 5 = 5\%$, $t = \frac{15}{5} = 3$
Due debt = 100 + 105 + 110 = 315

$\text{Installment} = \frac{94500}{315} \times 100$
= ₹30,000

242. (c) Cash payment = ₹19000
$\therefore$ Remaining amount
= 39000 – 19000 = ₹20000
Total amount paid as installment
= 5 × 4200 = ₹21000
$\therefore$ Total interest = 21000 – 20,000
= ₹1000
Let, rate of interest be R

$\therefore 1000 = \frac{(20000 + 15800 + 11600 + 7400 + 3200) \times R \times 1}{12 \times 100}$

$\Rightarrow 1000 = \frac{58000}{12 \times 100} R$

$\Rightarrow R = \frac{600}{29}\% = 20\frac{20}{29}\%$

243. (d) Principal = Rs.422092
R = 20%

$SI = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

Principal after 1 yr.
= 422092 – 21679 = Rs.400413

$\text{Amount} = \frac{422092 \times 20 \times 1}{100} + \frac{400413 \times 120 \times 1}{100}$

= 84418.4 + 480495.6 = 564914

244. (a) Principal after first year
= 141545 – 25490 = 116055

$SI = \frac{P \times R \times T}{100}$

$\text{Amount} = 1,41,545 \times \frac{11}{100} + 116055 \times \frac{111}{100}$

= 15569.95 + 128,821.05
= 1,44,391

245. (d) Principal = 623305
Rate = 10%
Time = 1 years

$SI = \frac{623305 \times 10 \times 1}{100} = 62330.5$

After I[st] year pranav return Rs.16390 of principal so principal for 2nd year
= 623305 – 16390 = 606915

$\text{SI of IInd year} = \frac{606915 \times 10 \times 1}{100}$

= 60691.5
Amount = 62330.5 + 606915 + 60691.5 = Rs.729937

246. (a) Principal = Rs.192208.
R = 15%

$SI = \frac{\text{Principal} \times \text{Time} \times \text{Rate}}{100}$

Principal after 1 year
= 192208 – 20196 = 172012
Amount pay end of second years

$= \frac{192208 \times 15 \times 1}{100} + \frac{172012 \times 115 \times 1}{100}$

= 226645
Note:- Using digital sum,
6 × 4 + 4 × 7 = 6 + 1 = 7
Option(a):- 226645
Digital sum of 226645 = 7

247. (b) Principal = Rs.598397,
R = 20%

$SI = \frac{P \times R \times T}{100}$

Principal after 1 year
= 598397 – 24284 = 574113

$\text{Amount} = \frac{598397 \times 20 \times 1}{100} + \frac{574113 \times 120 \times 1}{100} = 808615$

Note:- Using digital sum,
5 × 2 + 3 × 3 = 1 + 9 = 1
option (b) 808615
Digital sum of 808615 = 1

248. (b) Total amount = Rs.975
First installment = Rs.100
Time = n
2[nd] installment = 100 – 5 = Rs.95
So, the installment is in A.P
Now, using A.P sum formula
a = 100, n = ?
a_n = 975, d = –5

$S_n = \frac{n}{2}[2a + (n-1)d]$.....(i)

$\Rightarrow 975 = \frac{n}{2}[200 + (n-1) \times (-5)]$
$\Rightarrow 1950 = 200n - 5n^2 + 5n$
$\Rightarrow 1950 = 205n - 5n^2$
$\Rightarrow n^2 - 41n + 390 = 0$
$\Rightarrow n^2 - 26n - 15n + 390 = 0$
$\Rightarrow (n - 15)(n - 26) = 0$
$\Rightarrow n = 15$
$\therefore$ n = 26 is not satisfied in equation (i)
$\therefore$ Required no. of month = 15.

249.(b) Principal after first year
= 348020 – 15090 = 332930
ATQ,

Money paid = $3,48,020 \times \frac{8}{100} + 332930 \times \frac{108}{100}$

= 27,841.6 + 359,564.4 = 387,406

Note:-

To avoid calculation use digital sum method

$3,48,020 \times \frac{8}{100} + 332930 \times \frac{108}{100}$

= 8 × 8 + 2 × 9 = 1 + 9
= 1 (digital sum)

250.(c) Principal = Rs.335180
R = 13%

$SI = \frac{Principal \times Time \times Rate}{100}$

Principal after 1 yr.
= 335180 – 23060 = Rs.312120

$Amount = \frac{335180 \times 13 \times 1}{100} + \frac{312120 \times 113 \times 1}{100} = 396269$

Using digital sum method,
2 × 4 + 0 = 8
option (c) = 396269
Digital sum of 396269 = 8

251.(a) Interest = 250 + 310 – 550
= Rs. 10
Principal = 550 - 250 = 300

$Rate = \frac{SI \times 100}{Principal \times Time}$

$\therefore R = \frac{10 \times 100 \times 12}{300 \times 1} = 40\%$

252.(a) Interest = 25000 + 4 × 4000 – 40000 = Rs.1000
Remaining principal = 15000 + 11000 + 7000 + 3000 = 36000

$Rate = \frac{SI \times 100}{Principal \times time}$

$= \frac{1000 \times 100}{36000 \times \frac{1}{12}} = \frac{100}{3} = 33\frac{1}{3}\% \sim 33\%$

253. (c) Interest = 1340 + 1205 – 2540
= Rs.5

$Rate = \frac{SI \times 100}{Principal \times time}$

$= \frac{5 \times 100}{1200 \times \frac{1}{12}} = \frac{500 \times 12}{1200} = 5\%$

254.(d) Remaining amount = 2500 – 520 = 1980
Amount for installment = $1980 + 1980 \times 4 \times \frac{25}{1200} = 1980 + 165 = 2145$

$\therefore Installment = \frac{A \times 100}{100 \times T + \frac{RT(T-1)}{2}}$

$= \frac{214500}{400 + \frac{4}{2} \times \frac{25}{12} \times (3)}$

$= \frac{214500 \times 2}{825} = \frac{429000}{825} = ₹520$

Alternate Method:-

100
100 75
100 50
100 25
400 $\frac{150}{12}$ = 12.5

= 400 + 12.5 = 412.5
⇒ 412.5 → 2145

Monthly installment = $\frac{2145}{412.5} \times 100$
= 520

255.(a) Total paid interest after 11 month

$= 500000 \times \frac{3}{200} + 460000 \times \frac{3}{200} + 420000 \times .....$

$= \frac{10000 \times 3}{200}$ [50 + 46 + 42 + 38 + ...+10] = $150 \times \left(\frac{11}{2} \times 60\right)$

= 150 × 330 = 49500

Value of motorgages = Last term + Interest = 100000 + 49500
= 149500 ≈ 150000

Alternate Method:-

This method gives approx answer
Total installment amount
= 11 × 40,000 = 4,40,000
SI = 5,00,000 × 18% = 90,000
Value of motorgages
= 5,00,000 – 4,40,000 + 90,000
= 1,50,000

256. (b) Total Amount paid in installments
= 20 × 25000 = ₹ 5,00,000
Also, Rate = 10%
⇒ Amount on which interest is applied

$= \frac{500000}{11} \times 10 = 4,54,545.455$

∴ Down payment
= 6,50,000 – 454545.455 ≈ 1,95,455

257. (b) Principal = ₹ 1,00,000 , Rate = 10%

	Principal	Interest	Down payment
Ist year	1,00,000	10,000	10,000
IInd year	90,000	9,000	20,000
IIIrd year	70,000	7,000	30,000
IVth year	40,000	4,000	40,000

⇒ 5th year = 10,000 + 9,000 + 7,000 + 4,000 = ₹ 30,000

Alternate Method:

Total Principal = ₹100000
Paid in installment = 10000 + 20000 + 30000 + 40000 = ₹100000
Now in 5 th year he only pay the interest
= (100000 + 90000 + 70000 + 40000)

$\times \frac{10}{100} = 300000 \times \frac{1}{10} = ₹30000$

258. (b) Given P = 600 – 300 = 300

$S.I = \frac{300 \times r \times 2}{12 \times 100} \Rightarrow 60 = \frac{300 \times r \times 2}{12 \times 100}$

∴ r = 120%

259. (c) We have,
Due debt = ₹10192
Let the value of each installment
= 100 units
Due debt = 100 + 110 + 120 = 330
Value of each installment

$= \frac{10192}{330} \times 100 = ₹3088.48$



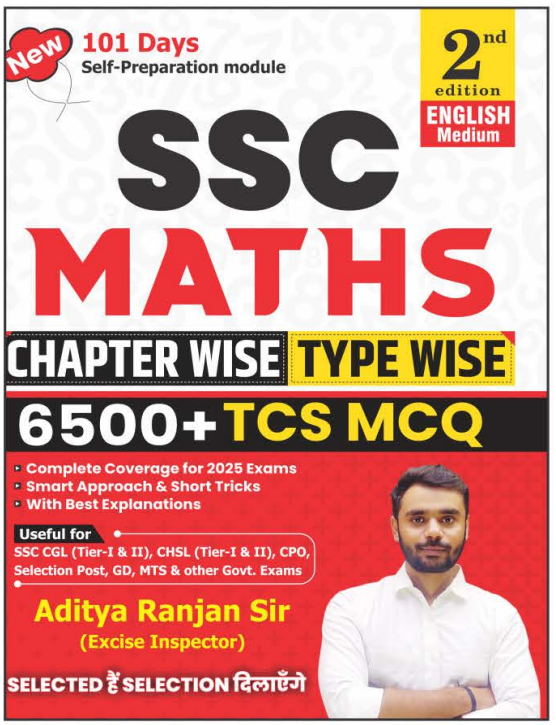
SPEED BOOSTER HACKS

♦ Some special results you should remember ♦

$(2)^1 = 2$	$(2)^6 = 64$	$(3)^1 = 3$	$(3)^6 = 729$	$(4)^1 = 4$	$(5)^1 = 5$
$(2)^2 = 4$	$(2)^7 = 128$	$(3)^2 = 9$	$(6)^1 = 6$	$(4)^2 = 16$	$(5)^2 = 25$
$(2)^3 = 8$	$(2)^8 = 256$	$(3)^3 = 27$	$(6)^2 = 36$	$(4)^3 = 64$	$(5)^3 = 125$
$(2)^4 = 16$	$(2)^9 = 512$	$(3)^4 = 81$	$(6)^3 = 216$	$(4)^4 = 256$	$(5)^4 = 625$
$(2)^5 = 32$	$(2)^{10} = 1024$	$(3)^5 = 243$	$(6)^4 = 1296$	$(4)^5 = 1024$	$(5)^5 = 3125$

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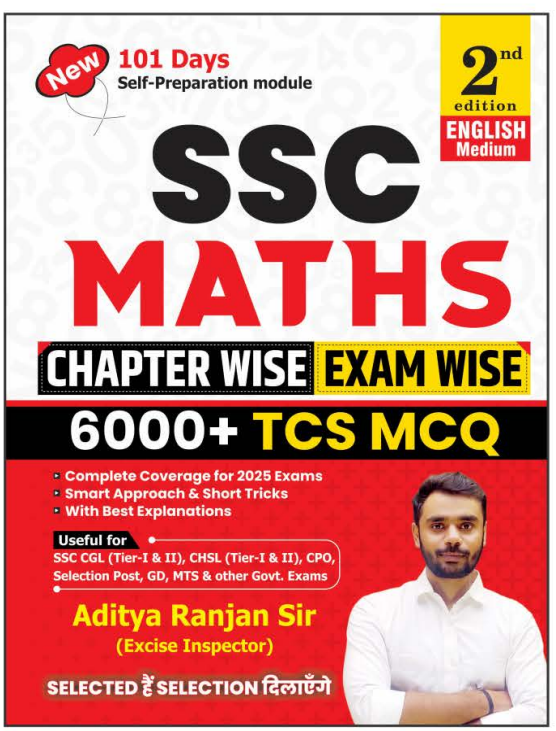
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